

Large Language Model-Based Multi-Agent Systems: Architectures, Collective Dynamics, and Emerging Frontiers

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1 Introduction

The emergence of large language model-based multi-agent systems (LLM-MAS) marks a significant confluence of two historically distinct research trajectories: the study of multi-agent systems (MAS) in classical distributed artificial intelligence, and the rapid advancement of large language models as cognitive engines. This convergence is not merely a technological juxtaposition but a fundamental reconfiguration of how intelligent behavior can be decomposed, distributed, and recomposed—a shift from the cognitive constraints of the individual model to the synergistic potential of an interacting collective [1, 2].

Classical multi-agent research established foundational frameworks for coordination, negotiation, and distributed problem-solving, primarily operating within well-defined, often symbolic or reinforcement learning-driven environments [3, 4]. While achieving success in constrained tasks such as warehouse automation and logistics, these systems were fundamentally limited by pre-defined communication protocols and a restricted capacity for open-ended reasoning [4]. Concurrently, LLMs have evolved from single-pass text generators to autonomous agents with profound abilities in planning, reasoning, and tool integration [5]. The strategic fusion of these fields addresses the linguistic and adaptability deficits of traditional MAS, envisioning a shift towards a division of cognitive labor where specialized agents cooperate through the universal interface of natural language [6].

The impetus for this paradigm shift extends beyond independent performance gains. Multi-agent composition serves as a powerful mechanism for mitigating the intrinsic limitations of monolithic LLMs such as hallucinations and finite context windows. By decomposing a complex objective into sub-tasks processed by specialized agents, systems can selectively mitigate individual model errors and extend cognitive capabilities beyond the constraints of any single context [7, 8]. More critically, this emergent collective intelligence does not always supersede individual baseline capabilities; instead, the rigorous orchestration of key principles within multi-agent systems, whether the central conductor or decentralized routing, becomes a determining factor for the emergence of genuine synergistic outcomes [9]. Yet, this composition is not a panacea; a systematic analysis of failures reveals that without careful orchestration, emergent problems of misalignment, cascading hallucination, and communication drift can introduce their own significant degradation of performance [7, 10].

The definitional boundaries of this field are established by the distinction from singular augmented agents and monolithic LLMs. While a single augmented agent integrates multiple tools to accomplish tasks sequentially, an LLM-MAS is an emergent collective of independent entities, each with a specific role and interacting through a communication topology such as peer-to-peer or hierarchical structure, where the aggregated behavior induces coordination dynamics and complex objectives [11]. This distinction has fundamental consequences for design. While recent empirical studies emphasize that the performance benefits of multi-agent collaboration often are conditional and can diminish as base LLMs improve, a deeper information-theoretic view reveals the interplay between task uncertainty, architectural choices, and communication in the field [12, 13]. Unlike traditional MAS, LLM-MAS harnesses the semantic priors and interactive reasoning capacities of LLMs to support a broad spectrum of collaboration strategies—from adversarial in debate to collaborative in code generation, and even competitive in market simulations [14].

The organization of this survey reflects the multifaceted challenges and opportunities inherent in the emergent field. We systematically dissect the field across several interlocking axes: the architectural designs that enable system cognition; the communication protocols and alignment of individual values that serve as the connective tissue of the system; the orchestration frameworks that manage the coordination of work; and the mechanisms that systems employ to learn, adapt, and self-improve. Our focus on evaluation (from cognitive microprocesses to systemic security) critically considers the research trajectory, moving past isolated success to adeptly

allocate attention in production. By foregrounding reliability across multiple levels—safety, human oversight, and the continuous interaction of models within the society—the survey offers a cohesive and scholarly map of the domain, effectively synthesizing the state of the art and guiding future investigations toward robust, controllable, and genuinely intelligent collective systems [15, 16].

2 Architectural Foundations and System Design

2.1 Individual Agent Design and Cognitive Architecture

The individual agent serves as the fundamental cognitive unit of LLM-based multi-agent systems, and its internal architecture—comprising planning mechanisms, memory structures, and tool integration—determines not only the agent’s standalone capability but also the upper bound of system-level performance. The prevailing architectural paradigm conceptualizes the LLM as the cognitive controller that orchestrates the agent’s internal components [1, 2], while the recent wave of design choices seek to more carefully balance the trade-offs between modularity, efficiency, and specialization [17].

Planning, Reasoning, and Decision-Making Modules. The reasoning capabilities of individual agents constitute the core differentiator between systems that merely distribute computation and those that genuinely leverage cognitive labor. Planning mechanisms are commonly instantiated through prompting techniques, including chain-of-thought for multi-step inference, self-consistency for variance reduction, and reflection-based frameworks for error correction and iterative improvement [2]. More sophisticated internal planning consists of hierarchical decomposition and dynamic re-planning, which makes agents responsive to environmental feedback and able to handle long-horizon objectives; this is the hallmark of agentic reasoning as distinct from single-shot inference [18]. However, the mere presence of such modules does not guarantee effective collaboration. Empirical studies reveal that in multi-agent settings, role-specialized agents exhibit notable performance differences on reasoning benchmarks, and reasoning performance is strongly influenced by the model choice and role assignment strategy [19]. Moreover, research on cognitive load theory suggests that when agents operate under excessive intrinsic or transactional load, the multiplicative burden of multi-step reasoning can degrade outcomes—highlighting the need for principled mechanisms that modulate cognitive demands in complex, multi-agent ecosystems [8]. These findings underscore the need for task-aware reasoning modules that adapt the depth, branching, and reflection frequency to the intrinsic complexity of the task rather than static reasoning templates.

Memory Systems Within an Agent. Memory in LLM-based agents spans multiple dimensions: sensory memory for immediate interaction context, working memory for maintaining task-relevant information across steps, and long-term memory for storing knowledge and personalization [20]. Within this framework, semantic memory (structured knowledge) and episodic memory (from past experiences) are distinguished from procedural knowledge, and operational processes—including consolidation, updating, indexing, mathematical and retrieval—are crucial for adapting to new information over time [20, 21]. Notably, empirical findings suggest that diverse implementations of memory strategies can lead to meaningful differences in system-level behavior and outcomes, though much of the existing memory architecture design remains heuristic [22]. A promising trend involves coupling agent’s individual memory with a shared system-level memory in a multi-agent architecture, creating a dual-layer memory hierarchy that supports individual learning and collective knowledge accumulation [11].

Integration of Tools and World Interaction. Extending the agent’s perception and action spaces through tools (APIs, code execution, web browsing, external knowledge) has become a standard pattern for modern LLM agents [1]. While a comprehensive market of tool-based agents is still emerging, we are seeing the emergence of frameworks that provide

well-defined abstraction layers and multimedia communication—such as protocol-level standards (e.g., TEA, MCP) that facilitate reproducible, versioned, end-to-end execution [23].

Critical Synthesis. A key consideration in designing individual agents for multi-agent systems is the locus of complexity: should the reasoning and memory reside entirely within each agent tailored entirely to the system-level. While intra-agent planning with iterative self-correction is a hallmark of robust individual agents, in multi-agent contexts it must be balanced against communication overhead and duplicated computation effort—points identified as failure patterns in MAS [7]. Moreover, the design of individual agents can be viewed as a configuration in a larger network topology—rather than solitary intelligent authors, agents are interlinking through reasoning pipelines whose individual design choices interact with the global orchestration to produce system-level outcomes [24]. This observation suggests that individual agent design should be jointly optimized with communication topologies, and that successful methods will leverage heterogeneous strengthening models with role-fine-tuned components [25].

2.2 Communication Protocols and Interaction Interfaces

The interaction substrate of an LLM-based multi-agent system determines how the cognitive capacities described in the previous section are actually brought to bear on shared problems, governing not only what information is exchanged but also the reliability, cost, and interpretability of the emergent problem-solving process. Indeed, the internal memory hierarchies and tool interfaces of individual agents constitute only one layer of computational infrastructure: memory systems may be shared, and tools may be accessed by multiple agents, making the communication substrate the critical bridge that connects individual cognition to collective action. The field has evolved through three primary communication paradigms—natural language, structured protocols, and latent-space transmission—each representing a distinct trade-off between expressivity, efficiency, and control. This design decision is fundamental: the choice of communication protocol profoundly shapes the system’s semantic fidelity, scalability, and vulnerability to cascading errors [26]. Crucially, this substrate forms the scaffolding upon which the orchestration layer constructs its control regimes: the expressiveness or restrictiveness of the chosen communication medium directly constrains which coordination structures are viable, establishing the boundary conditions within which role assignment, task decomposition, and dynamic control—the subjects of the next subsection—can operate.

The predominant paradigm is natural language messaging, which leverages the inherent linguistic competence of LLMs to enable direct, human-comprehensible interactions without elaborate encodings [27]. This interpretability is invaluable for debugging and human oversight, and research shows that agents communicating in natural language can earn more trust and cooperate more effectively with humans [27]. However, the same open-ended flexibility that makes this paradigm so accessible also introduces serious scalability bottlenecks. The per-message token cost drives significant computational overhead, while the inherent ambiguity of unstructured language makes semantic drift and error propagation more likely in long-horizon tasks—a critical concern, since much of the reasoning that unfolds between agents can be understood as successive stages of incomplete rationalization. This dual nature is a core issue, since the interpretability and trust it confers mask a deep-seated fragility.

To mitigate these limitations of open-ended language, the field is developing more structured and standardized protocols. A key development is the recognition of context-oriented protocols (governing tool/agent-environment interaction) versus inter-agent protocols, leading to standardized frameworks such as the Tool-Environment-Agent (TEA) protocol and the Model Context Protocol (MCP) to define interactions [23, 28]. It is critical to design these and other protocols across the layers involved—while the *Web of Agents* position shows that shifting semantics from "data" (Semantic Web) to "models" (LLM-era) has displaced the focus from infrastructure to the agent’s reasoning [29]. These standards provide a formal structure that improves reliability, enables provenance and version control, and—critically—provides the semantic hooks that orchestration

frameworks need in order to route messages, enforce role boundaries, and maintain the system’s architectural integrity. The end goal is the construction of a whole ecosystem of humans, agents, and tools connected through the Web of Agents, which will require robust, layered communication infrastructure that is both secure and efficient across these divergent protocols.

The most advanced trend in this space is representing communication as internal, structured, reusable latent spaces. Techniques like *Agent Primitives* [30] impose constraints on multi-agent interactions—moving beyond raw text to use KV-cache and other compressed state representations for state transfer—to mitigate information degradation across long-context stages while improving efficiency. This paradigm allows for a more efficient transmission distribution and aligns with a broader design insight: that the capabilities of current LLM agents are increasingly harnessed and externalized into computational structures that often exceed the cognitive processes they formerly represented [31]. Such latent-state communication transfers not just messages but also serialized reasoning states across system boundaries, allowing heterogeneous components to be coordinated in ways not bound to rigid surface linguistic form.

Although the three major interaction paradigms are individually well-suited to specific tasks, the current direction of study is strongly focused on pragmatic interoperability and multi-paradigm coexistence. Several works call for the development of more mature design patterns inspired by software engineering—e.g., Mediator, Observer, and Publish-Subscribe—as the intellectual scaffolding for communication between LLM-based agents [28]. This is accompanied by a pressing need for the ecosystem to solve the "missing layer" problem whereby protocols must connect the full lifecycle support of tools and agents within the shared infrastructure [23]. Future progress depends less on protocol design alone than on addressing the interoperability of standards, ensuring security, provenance, and latency guarantees, and addressing the need to drive durable, long-running processes within these distributed systems to improve capabilities over time [26, 29]. The next subsection will explore how these constraints feed into the orchestration layer, and how the specific topology and coordination of roles place further requirements on the communication substrate described here—requirements that must be met dynamically without sacrificing reliability.

2.3 Coordination, Orchestration, and System-Level Architecture

The orchestration of LLM-based agents constitutes the architectural "brain" that transforms a collection of independently capable models into a cohesive, goal-directed system. This coordination layer is fundamentally concerned with three intertwined problems: which agents should perform which roles, how tasks should be decomposed and sequenced, and how control should be distributed across the system.

Agent selection and role specialization establish the foundational division of cognitive labor. Approaches span from homogeneous pools of interchangeable models that decompose problems through simple parallelization, to heterogeneous systems where specialized LLMs are composed based on task-relevant capabilities [32]. Dynamic role assignment is increasingly viable, where the controller evaluates task features and constructs coalitions of complementary agents on the fly, addressing the challenge that static configurations frequently misallocate cognitive resources to tasks of varying complexity [33]. Graph-based methods such as Graph-of-Agents further the selection process by constructing communication structures based on each model’s demonstrated capabilities, enabling parsimonious pools of agents to exceed the performance of larger, undifferentiated collectives [34].

Task decomposition translates an abstract objective into a set of interlocking, executable units. The structure of this decomposition—sequential chains, hierarchical trees, star arrangements, or general directed acyclic graphs—determines not only information flow but also the system’s resilience to error propagation [35]. The formalism of task graphs facilitates the construction of control networks, where the choice of topology critically affects both accuracy and operational characteristics. Empirical studies of the information-propagation effects reveal a nuanced

relationship: moderately sparse topologies often outperform both dense and extremely sparse structures by suppressing error amplification while preserving necessary broadcast of credible information, directly quantifying the trade-off between efficiency and systemic stability [35].

Orchestration frameworks embody the control regimes that govern these tasks. Centralized frameworks, featuring a distinct conductor or planner, simplify task assignment and conflict resolution but create at-risk points and potential bottlenecks; they often follow a static workflow that is susceptible to misalignment and partial information. Hierarchical designs mitigate these risks by enabling meta-level decomposition into sub-teams, while fully decentralized structures are more robust but can lead to resource redundancy and coordination failures [14]. The design space extends to dynamic orchestration, where the system’s arrangement itself—its decision hierarchy, information-flow routing, and role assignments—must adapt in response to task feedback and emerging dynamics. This problem of topology learning as an emerging computational problem has driven attempts at dynamic graph construction, with generative approaches in particular employing graph diffusion models to generate task-adaptive sparse topologies in real time based on a proxy measurement of the multi-objective reward function [36]. Similarly, trust-based mechanisms inject a first-class notion of message-level assurance into the system to filter unwanted propagation and allow for the rerouting of computational services in adverse settings [37].

The orchestration framework ultimately exerts a decisive influence on the system’s security surface. Centralized architectures consolidate lines of communication, making them targets for latency and bottleneck attacks, whereas decentralized magnified surfaces solicit greater exposure to broader attack vectors. Therefore, secure orchestration cannot be treated as purely functional arrangement with respect to maliciously spoofed communication topologies [38]. A promising direction lies in problem-informed topologies where communication structure mirrors the inherent logical decomposition of the domain, providing a way to balance content-based and feature-based constraints. The current state guarantees that orchestration architecture must be considered a first-class independent variable—the strategic repertoire enabling coordination in the space of model quality, security, and cost.

2.4 Cross-System Resource Management and Collective Memory

As multi-agent systems scale from handfuls of specialized collaborators to ecosystems of dozens or hundreds of interacting models, the management of shared computational and informational resources becomes a first-order architectural concern. While the preceding section addressed the structural orchestration of agents—their roles, task decomposition, and control regimes—this subsection examines the **systemic substrate**: the collective memory, information routing protocols, and economic schedulers that determine whether such systems operate with coherence and efficiency or degrade into redundant, inconsistent, and budget-exhausting processes. This resource layer directly mediates the practical viability of any orchestration strategy, and in turn, establishes the operational constraints—cost, latency, and reproducibility—that the following section on systems engineering must confront.

The foundation of collective knowledge in multi-agent systems rests on shared memory infrastructures that persist beyond individual agent context windows. These systems are increasingly organized as **three-layer hierarchies**, conceptually analogous to computer architecture: I/O buffers for immediate task context, caches for frequently accessed shared knowledge, and persistent memory stores for long-term consolidation [39]. Within this hierarchy, shared memory frameworks are typically partitioned to enforce information segregation, addressing the critical challenge that not all agents should access all knowledge. This is particularly consequential in cross-organizational deployments where privacy concerns necessitate decentralized coordination with minimal centralized data exchange [11]. The integration of procedural memory—decomposed into reusable units and flexibly allocated across orchestrators and task agents—further illustrates the architectural depth of shared memory, mirroring the hierarchical decompositions between

orchestrators and workers, and enabling smaller models to approach the performance of stronger backbones via fine-grained memory capabilities [40]. Critically, orchestrator-level memory is essential for effective task decomposition, while agent-level memory empowers smaller, cheaper models to close the performance gap to larger backbones, thereby impacting resource allocation [40].

However, a persistent and under-addressed challenge in multi-agent memory is **consistency preservation** [39]. When LLMs read and write shared memory in parallel, divergent knowledge states can emerge, and without proper transaction control, task execution can suffer from uncoordinated context updates. SagaLLM addresses this by adapting the Saga transaction pattern—originally from distributed databases—to LLM planning, incorporating managed execution and compensable actions to ensure recoverability and constraint satisfaction across multi-agent workflows [41]. This borrowing of systems-level guarantees is a direct precursor to the operating-system-inspired management that the following subsection will treat as a central engineering principle.

Beyond memory, the efficient operation of large-scale multi-agent systems hinges on systematic information management and scheduling. Rather than uniformly transmitting full trajectories, advanced frameworks employ **dynamic task graphs and routing mechanisms** to control data flow, adaptively orchestrating asynchronous and parallel execution [42]. Similarly, compressing previous interactions—extracting only essential facts into a condensed context—can streamline multi-agent coordination and reduce the token burden [43]. This per-query routing logic manifests in systems like MasRouter, which learns to determine whether collaboration is necessary and, if so, selects both the optimal collaboration mode and the most suitable agents, achieving a 17–28% overhead reduction versus fixed collaboration structures [25], implicitly adjusting the orchestration topology on the fly.

Effective orchestration, however, depends on the available economic **budget**, linking information access to financial allocation. Modern frameworks introduce observer roles that autonomously refine plans and strategies, mitigating premature redundancy and controlling resource expansion [44]. More principled approaches formalize this as a utility-guided orchestration problem, weighing estimated gain against step cost, uncertainty, and redundancy to minimize the expected cost [45]. At the system-design level, resource allocation is framed as an optimization problem under a strict budget, determining which agents are activated and in what structure—a supernet search of agent architectures [46]. This birth of a centralized scheduler realizes the CPU-like role in the system, controlling the lifecycles of dynamic task graphs and adapting to asynchronous decision-making [42].

In synthesizing these threads, a clear frontier emerges: the governance of informational and computational assets transitions from a design-time concern to a runtime, adaptive discipline. Ensuring robustness in these operational environments requires moving beyond static context windows to a **systemic memory**—one that integrates tools, execution states, and capabilities with dynamic resource awareness [39]. The next generation of orchestration frameworks must handle probabilistic resource consumption and hard deployment constraints with the agility of an operating system managing its processes—a parallel that the following subsection will develop into a formal design philosophy, bridging the conceptual architecture of these systems with their reproducible, engineered reality.

2.5 Engineering Practices, Scalability, and Design Formalization

The pragmatic realization of LLM-based multi-agent systems (MAS) hinges on a triad of engineering concerns: the availability of robust development frameworks, the formalization of design principles, and an understanding of scaling behavior. While earlier subsections established the conceptual building blocks, this section grounds those abstractions in the practical realities of construction, evaluation, and deployment, addressing the critical gap between architectural vision and operational viability.

A wave of orchestration frameworks has sought to democratize MAS development. MASLab stands out as a unified codebase consolidating more than twenty established methods, enabling the critical component of fair, head-to-head comparison of orchestration strategies, which is often obscured by heterogeneous implementations [47]. The need for methodological control is further underscored by the analysis of systems like MAESTRO, which provides a standardized execution environment that decouples architectural design choices from backend model variability. Its empirical findings that the structural composition of a system is the dominant driver of its reproducibility and cost-over-performance trade-off is a crucial data point for designers, asserting that topology is an optimization target in its own right, irrespective of the underlying model’s strength [48]. To manage the complexity of heterogeneous agents and tooling, protocols such as the Tool-Environment-Agent (TEA) protocol have emerged to treat these components as first-class, versioned resources with explicit lifecycles, addressing the brittleness of ad-hoc integrations and enhancing traceability [23]. This theme of lifecycle management extends to self-evolving protocols like Autogenesis, which formalizes versioned interfaces to allow for auditable, closed-loop self-improvement without breaking dependent components [49].

Complementing these practical frameworks is a need for theoretical formalization to move beyond anecdotal success. The borrowing of the Saga transaction pattern provides a powerful analogy for LLM planning, incorporating checkpointing, compensation, and independent validation agents to provide durability and recovery in distributed LLM workflows [41]. Similarly, the application of operating system principles is not just metaphorical. The design of resource managers such as the AgentRM, which applies concepts like Multi-Level Feedback Queue scheduling and context lifecycle management to prevent zombie processes and memory bloat, demonstrates that decades of systems-engineering wisdom are directly transferable to stabilize agent loops [50]. This cross-pollination suggests a new era of "agent systems engineering" where latent middleware manages agent autonomy in the same way OS manages resources.

Perhaps the most contested area is the empirical study of scaling dynamics. Counter-intuitive results challenge the orthodoxy of "more is better." In controlled studies of multi-hop reasoning under equal token budgets, single agents can match or outperform their multi-agent counterparts, suggesting that some reported MAS gains are artifacts of unaccounted test-time computation rather than emergent collaboration, highlighting the importance of explicitly controlling compute, context, and coordination [51]. With the corroboration of these findings, the rethinking of homogeneous workflows demonstrates that a single agent, with KV cache reuse, can replicate the performance of prompt-based multi-agent workflows at a fraction of the economic cost, positioning this single-LLM implementation as an important baseline [52]. This is the core of the Agent System Trilemma: balancing performance, cost, and latency [53] — additionally, it is critical to consider training-based efficiency approaches like Optima, which substantially improves both communication efficiency and task effectiveness, yielding performance gain while markedly reducing token consumption [54]. Rather than merely increasing agent counts, EvoRoute focuses on dynamic routing of requests by selecting Pareto-optimal LLM backbones—an important shift from naive aggregation to more intelligent resource orchestration, achieving up to 80% cost reduction while maintaining or enhancing performance [53]. Systems designed with efficiency in mind, such as MasRouter, formalize this paradigm by integrating collaboration mode and role allocation into a unified routing problem, optimizing the entire MAS selection pipeline as a coherent unit [25].

In conclusion, the new engineering of LLM-MAS is dynamically steering away from naive scaling and primitive ad-hoc orchestration. The next frontier lies in the quantitative formalization of optimal communication topology (e.g., EIB-learner balancing error suppression and beneficial information propagation) [35] and in the development of self-evolving systems that operate under strict and transparent constraints. The future will be won by synthesis—integrating the structured realities of OS-like management with the dynamic adaptability of self-organizing systems, all while providing rigorous scientific validation via controlled, unbiased codebases

such as MASLab to ensure the field progresses based on reproducibility of robust engineering outcomes rather than anecdote insights.

3 Collaboration, Communication, and Collective Decision-Making

3.1 Deliberative Mechanisms and Debate Protocols

Deliberative mechanisms constitute the procedural backbone of LLM-based multi-agent systems, transforming isolated model outputs into collectively reasoned conclusions through structured argument exchange and iterative refinement. Unlike static ensemble methods that naively aggregate independent predictions, debate protocols create an interactive epistemic environment in which proposals are subjected to scrutiny, critique, and revision before final consolidation. The design space for these mechanisms spans a critical continuum from competitive, adversarial configurations—where agents adopt opposing stances to stress-test a candidate solution—to consensus-seeking frameworks designed to promote convergence through structured negotiation [6, 55].

Within competitive debate architectures, the Mixture-of-Agents (MoA) paradigm has established a foundational layered structure in which agents at each level incorporate outputs from the previous layer as auxiliary context, achieving substantial performance gains over strong single-model baselines [56]. More recent work has systematically formalized these approaches, observing that multi-agent debate systems fundamentally work by aggregating adequate information across diverse agent perspectives, representing a departure from pure RL fine-tuning or majority voting [57]. However, the defining challenge is the aggregation protocol: how individual positions are fused into a single collective output. Voting schemes are computationally simple and maximally scalable, but they privilege prevalence over quality, rewarding output richness at the expense of signal fidelity when the majority is misled. Adversarial adjudication introduces a distinct "critic" agent tasked with evaluative synthesis, but raises questions about the adjudicator's own reliability and potential single-point-of-failure vulnerabilities. Bayesian belief updating and confidence-weighted averaging offer principled probabilistic grounding, yet they require reliable confidence calibration, and LLMs exhibit marked miscalibration, particularly under domain-shift and misleading priors [51]. This tension underscores a broader coordination-versus-task-alignment principle: the optimal number of agents depends on the intrinsic difficulty of the task, its decomposability, and the baseline capability of the individual agents involved [58].

The efficacy of any deliberative system is bounded by its ability to avoid collapse. A well-established pathology is *over-deliberation drift*, where extended rounds of exchange amplify initially minor errors rather than correcting them: sustained iterative refinement monotonically improves outputs only up to a saturation point, beyond which performance degrades significantly because agents increasingly anchor to prior outputs, creating echo-chamber-like dynamics and diminishing the value of additional exchanges [59]. Sycophancy-induced convergence presents an orthogonal failure mode: critical evaluation becomes shallow and perfunctory as agents aim to agree with their peers, particularly when the system lacks strong accountability or when expert-backed proposals are not given sufficient weight [10]. This influence asymmetry ties deliberation failures to structural heterogeneity. Emergent coordination studies reveal that agent heterogeneity is crucial—homogeneous pools saturate early, while diverse agents contribute complementary evidence, improving effective collective deliberation [58]. Yet, when agents fail to perform effective role differentiation, skilled agents are not leveraged properly and teams regress from optimal performance [10]. Critically, empirical evidence shows that strong single-agent reasoning performance bottlenecks the potential benefits of multi-agent debate systems; when single-agent baselines exceed a threshold, the marginal performance benefit from additional coordination diminishes dramatically or becomes negative [51, 13]. These findings collectively point to a critical design condition: agent count alone is insufficient, and the deliberative system

must be calibrated to the capability distribution of its constituent agent pool.

To maintain productive deliberation, modern frameworks increasingly implement adaptive governance. This involves three core mechanisms: dynamic control of information flow, uncertainty-aware stopping conditions, and deliberate engineering of agent diversity. Consensus-seeking protocols that decouple information exchange from rigidity have proven effective at maintaining information richness in deliberative contexts [55]. Resource-aware controllers, such as the Conductor, learn via RL to orchestrate collaborative interactions dynamically—designing communication topologies and structured prompts on the fly—instead of relying on static pre-configurations [60]. This "puppeteer-style" paradigm—where a centralized orchestrator issues sequential and prioritized communication directives, trained end-to-end to optimize system outcome—demonstrates that adaptive conversational structure yields improved performance at reduced computational costs [9]. Failure-aware evaluation frameworks (e.g., MAST) reveal that the majority of failures in multi-agent systems arise from *inter-agent misalignment* and *task verification* failures, and that such failures are highly robust to naive mitigation, instead requiring principled design changes [7]. These findings underscore that **debate quality degrades with decentralized, uncontrolled peer-review** and improves with institutionalized role differentiation and adaptive control.

Looking forward, the open frontier for deliberative mechanisms is in *learnable confidence-aware orchestration*: deciding when to deliberate deeply, when to limit discussion, and when to defer to a single agent. Existing work shows that heterogeneous MAS configurations—mixing different model backbones—can dramatically outperform homogeneous teams [19]. Further integration of formal Bayesian frameworks could make referee decisions principled and well-calibrated under uncertainty. The field is moving from static multi-agent workflows built for the worst case toward adaptive and context-aware deliberations, where debate is invoked only when it is expected to yield positive epistemic benefit.

3.2 Incentive Structures and Strategic Interaction Dynamics

Incentive structures constitute the invisible hand that governs multi-agent systems, shaping whether collections of LLMs converge toward cooperative problem-solving, degenerate into competitive opportunism, or exhibit pathological coordination failures. While the preceding discussion has established how *who speaks, when, and how much weight each voice carries* constitutes a first-order architectural lever on deliberative quality, the incentive mechanisms examined here address a deeper question: *why* agents choose to participate productively at all, and how the formal and informal reward landscapes can be engineered to sustain that willingness over extended interactions. Unlike classical multi-agent systems studied within game theory—where utility functions are explicitly designed and globally known—LLM-based systems present a fundamentally more complex design problem: agent preferences emerge from elicited prompting and model internals, are negotiated through natural language rather than formal protocols, and evolve dynamically as agents infer each other’s beliefs and intentions during interaction. The design of incentive structures thus operates at two coupled levels: the formal specification of reward mechanisms, task allocation, and outcome evaluation that govern system-level objectives, and the emergent, socially constructed incentive landscape arising from the rhetorical interplay of agents attempting to influence and predict one another [61]. This dual-layered view directly extends the deliberative design space explored previously—where voting rules and debate protocols de facto determine whose voice counts—by asking what causes agents to voice informative positions in the first place.

Methodologically, this section emphasizes design over architecture. While debate and orchestration protocols constitute the formal skeleton of emergent coordination, incentives constitute its functional musculature—they determine whether the coordination that emerges is productive or pathological. Understanding the incentive space requires examining the "invisible key-value pairs" that connect individual agent behavior to collective system outcomes.

The foundational challenge is aligning individual agent objectives with system-level behavior. Topological coordination and communication structure serve as a primary incentive lever: coordination topologies fundamentally establish the scope of behaviors by altering what agents can observe and anticipate [13]. Systematic exploration reveals that homogeneous agent pools display strong diminishing returns as agent count increases, while diversified composition mitigates this effect—the diversity of independent informational channels constitutes a structural determinant of distributed system performance [58]. This structural insight is operationalized in frameworks such as SYMPHONY, where heterogeneous LLM assemblies generate exploration diversity during tree-search-based planning [62], and in agent primitives that communicate via key-value caches, making agent artifacts usable and structured [30]. These results independently corroborate the deliberative analysis of the previous section—where heterogeneous contributor pools evaded echo-chamber collapse—with formal information-theoretic language applicable both to deliberative efficacy and strategic incentive design. The efficient allocation of agents to roles thus becomes a central design lever: frameworks automating agent search use qualitative design spaces [63], while others tie agent incentives to explicit sub-task solvability, completeness, and non-redundancy constraints [64]. Multi-agent architectures also embed incentives directly: hierarchical meta-cognitive structures enable agents to model, monitor, and control an "executive" process, setting "meta" as an executive agent over lower-level execution agents during complex reasoning—incentivizing containment of reasoning errors and iterative refinement within a single architectural scaffold [65].

While these structural incentives shape *ex ante* behavior, LLM agents' strategic dynamics also emerge from the dynamics of their interactions, particularly in state-of-the-art situations. Explicit reasoning about other agents' beliefs and strategies—emblematic of Theory of Mind (ToM)—qualitatively changes decision outcomes [66]. The key point is that the interplay of mental models and strategic interaction can be understood as game-theoretic equilibria: Bayesian Nash equilibrium-seeking approaches, where each agent computes a personal optimal response subject to beliefs about others' strategies, enable rational convergence without costly multi-agent exchange [67]. These insights parallel the incentive pathologies of deliberative systems discussed previously—echo-chamber amplification and sycophancy collapse—while showing that equilibrium-based reasoning can replace move-level deliberation with belief-level convergence, trading interaction cost for computational reasoning cost. Complementary, self-play-based co-optimization methods allow agents to reason through emergent strategic interaction, though they expose the bottleneck of turn-level credit assignment in multi-agent RL settings [68].

Beneath these strategic dynamics resides a critical structural intuition rarely rendered explicit in multi-agent systems: incentives are not a modifiable add-on but are embedded in the design space itself. The incentive structures of LLM-driven multi-agent systems are, in their specificity, closer to the executable (system-level "scripts" that ensure non-redundancy of sub-managerial tasked actions) and to the emergent (dynamics of agent-game dynamics).

Importantly, from the assessment of strategic incentive mechanisms, we transition to examining the surveillance of long-range and memory-mediated incentive controls—a natural bridge to what follows next.

Behavioral pathologies. Agent behavioral "drift" is again a long-term effect: semantic, coordination, and behavioral drift cause progressive degradation [69]. Additionally, when the system processes adversarial and strategic requests, current LLM agents suppress either positive or negative risk evidence, indicating a tendency to trust memory or authoritative sources are above the system's strategic objectives [70]. Adaptive memory and memory structure selection (optimized as an objective) can be a key determinant of strategy behavior: a well-managed memory is used as an effective incentive for consistent future behavior [71, 71]. This enables the critical insight that incentives are revealed not solely in protocol states but embedded in the records or preferences—an underappreciated vulnerability, as attacks such as memory-control-flow highlight [72]. These insights reinforce the previous section's emphasis on adaptive

coordinating agents: governance of memory and information flow—be it in debate dynamics or strategic incentives—is an operational requirement as much as an optimization.

The preceding discussion has emphasized structural and emergent incentives, but a third, and often overlooked, cornerstone of incentive design is: **the cost of coordination**. Deliberative subsystem quality decays with unconstrained interaction—but in this context, "unconstrained" means **without incentive-aware budget management**. Strong single-agent baselines show that many homogeneous multi-agent workflows can be implemented by one agent reliably, without coordination and often yielding comparable performance [52]. This places the incentive design question at the center: coordinating toward single-agent already includes utility maximization—then is a multi-agent system always the optimal? From the managerial-economics standpoint, agents are not only reward-driven but **budget-constrained**. The incentive question is then not simply "how to align objectives" but "when does alignment justify the overhead of coordination?"—which directly implies the optimality condition between coordination overhead and incremental benefits.

The efficient optimal (from "selfish" standpoint) allocation of multi-agent resources—the transposition of **several components** in this framework—is again reinforced by the insight that distributed configurations that minimize coordination overhead are those that structure cooperation along natural boundaries, leveraging unique per-agent contributions [8]. In concert with the observation that the best distributed systems avoid over-optimizing **behavioral** interaction, this yields the practical rule-of-thumb for agent system designers: **incentive design granularity, including reward granularity and distribution, may be as important as agent quantity**.

A final, emerging frontier is the **self-evolution of the incentive system** itself, where meta-level techniques modify the incentive design as a function of system behavior. Low-fidelity memory architectures lead to increased duplication of tasks, increased inefficiency; high-skill laziness (strategic with capacity but low will) potentially leads to collusion and cost [73]. Future research on calibrating "good" incentive strategies through memory governance and periodic memory consolidation [71] will likely grow. At the meta level, evolutionary optimization of communication topologies and the agent-design prompt space [24, 46] directly treats the "incentive set" itself as a search variable, and concurrently guides the evolutionary "incentive set" from a static design choice to a dynamic, optimized-parameter—relying on the frameworks developed for runtime monitoring and active interventions [74]. As multi-agent systems transition to production, incentive-driven design is rapidly evolving into an engineering discipline that optimizes both explicit and implicit reward mechanisms, integrated with architectural learning [65, 72]—moving the field towards systems capable of sustaining the high costs of collaboration in real-world deployments.

The above structural and strategic incentive architectures—procedural, incentive-based, contingent over the adaptive environment—point towards a unified principle: at all levels, the fading of the "incentive alignment" is not a special challenge but part of what makes multi-agent collaboration stick. The *optimal* level of control is not a fixed number but a coefficient of the "economic exchange rate" of the coordination cost business environment. The upcoming section will guarantee this principle: **task decomposition, role specialization, and planning/execution mechanics** will provide the operational core where the incentive structures analyzed here are operationalized as concrete computational tasks. The incentive and procedural "productive friction" interaction is the core research object of the remainder of this survey.

3.3 Task Decomposition, Roles, and Multi-Agent Planning

The decomposition of complex objectives into tractable sub-tasks, the assignment of specialized roles, and the orchestration of these components through structured plans constitute the operational core of LLM-based multi-agent systems. This procedural layer determines not

merely efficiency but the fundamental feasibility of collective problem-solving; the manner in which a problem is partitioned dictates the extent to which specialized agents can contribute meaningfully without redundancy or conflict [28, 26]. Classical formulations such as modular interpreted systems provide formal grounding for synchronous, modular multi-agent modeling, yet LLM-based systems demand more flexible, semantics-aware decomposition that can adapt to the open-ended nature of natural language tasks [75].

Constraint-based decomposition formalizes task division through dependency maintenance, ensuring that logical and temporal precedence relationships persist across sub-tasks. These approaches treat abstract task representations as constraint satisfaction problems, where modularity, coupling, and critical-path characteristics can be assessed a priori. Effective role allocation, in turn, converts such decompositions into actionable assignments. The Multi-Agent Collaboration Mechanisms survey delineates role-based strategies as a key dimension of collaboration, where agents are endowed with differentiated capabilities whose relevance to tasks is evaluated to form coalitions under resource constraints [14]. This capability-aware assignment mitigates the monoculture problem in homogeneous pools, where system-level behavioral variance collapses, while dynamic role reconfiguration—where ascriptions shift according to evolving task states—enables adaptation without complete architectural re-specification [32].

Dynamic task graph generation extends this decomposition logic into adaptive workflow construction, leveraging hierarchical planning and iterative decomposition structures. These systems typically leverage a "planner-executor" architecture: a control agent generates a sequence of sub-goals, guided by local feedback and global constraints, then dispatches them to specialist agents whose outputs are collected and validated. The generative topology of such graphs remains a pressing open challenge; static, hand-engineered workflows are brittle across task distributions, and recent diffusion-based approaches (e.g., Guided Topology Diffusion) attempt to synthesize task-adaptive structures that optimize for communication cost, robustness, and accuracy simultaneously [36]. Existing distributed systems exhibit "topology-dependent" performance degradation during perturbations—sparser but regular graphs preserve cooperative accuracy, while dense graphs are less robust to cascading failures when injecting errors [35].

Role allocation and planning are simultaneously restricted and enabled by the original communication infrastructure. From standardized formats like A2A to hybrid and latent-space systems, the inter-agent communication stack defines both task granularity and the expressive capacity for problem-solving [76, 77]. For instance, a task-driven role allocation in healthcare environments, or resource-constrained role formation tasks, exposes the limitations of pure global or distributed planners, and there is growing pressure to "marry" structure and semantics in agent planning formalisms [33, 30]. Hierarchical planning, in particular, provides a algorithmic answer: decomposing down the hierarchy decreases control complexity and communication distance; failure to do so leads to semantic "maintenance" in centralized roles.

Multi-agent planning under uncertainty, in its most practical form, attempts to bridge the gap between an idealized static plan and the exigencies of execution. Feedback-driven execution, such as incorporating direct answer validation and inter-agent referencing, ensures plan execution remains aligned with the original objectives despite variable intermediate outputs [78]. However, existing coordination often ignores the cumulative cost. Design ablation studies have shown that multi-agent systems can decrease performance due to "coordination overhead" unless disciplined by an explicit orchestrator, and that the practical reality of LLM delegation is more brittle than assumed—inter-agent delegation chains corrupt subtly during execution [79]. As topologies become more adaptive, they must explicitly distinguish between "expressive but expensive" and "efficient yet tightly controlled" regimes, a distinction likely to dominate performance and reliability frontiers in recent deployments.

3.4 Social Cognition, Perspective-Taking, and Interactive Decision-Making

Social cognition constitutes a defining frontier in LLM-based multi-agent systems, distinguishing mere communication from genuine collaborative intelligence. While the preceding discussion has examined how procedural mechanisms—task decomposition, role allocation, and communication architectures—channel agent behaviors into executable workflows, this subsection addresses a complementary question: what enables agents to move beyond executing predefined roles toward genuinely adaptive, anticipatory interaction? The capacity to represent, reason about, and predict the mental states of interacting peers—collectively termed Theory of Mind (ToM)—underpins sophisticated interactive decision-making across cooperative, competitive, and mixed-motive settings [80]. This cognitive machinery enables agents to move beyond reactive exchanges toward anticipatory strategic behavior that accounts for beliefs, intentions, and knowledge asymmetries of their counterparts. Crucially, the depth of ToM reasoning an agent can exercise is bounded by the autonomy and flexibility afforded by its assigned role and the granularity of task decomposition—the architectural mechanisms examined earlier—while simultaneously determining the quality of human-agent interaction explored in the following subsection: whether an agent can recognize when a human overseer is confused, uncertain, or misinformed depends directly on its capacity for mental-state attribution.

The central computational challenge in multi-agent ToM is the recursive modeling of belief states: an agent must reason not only about what others know, but also about what others believe the agent itself knows. This requires adaptive estimators that calibrate the depth of recursive reasoning to observed behavior of partners, avoiding both under-mentalizing (treating other agents as purely reactive responders) and over-mentalizing (attributing unrealistic strategic depth that degrades system performance). The design of such estimators involves a critical trade-off between computational overhead and predictive fidelity. In interaction-centric benchmarks, analyses of emergent coordination reveal structurally observable differences in whether networks of agents display higher-order collectives or merely individual aggregates, with synergistic evidence measurable through information-gain-based emergence criteria and content-differentiation metrics [81]. This suggests that effective ToM alignment is not merely an architectural feature but dynamic process of group formation in which attribution emerges from interaction patterns—a finding that directly complements the dynamic task graph generation discussed previously, as both emphasize adaptive structure over static design.

Formal models of strategic thinking extend foundational ToM toward more comprehensive decision-theoretic frameworks. Agents engaged in interactive decision-making must solve tasks that incorporate iterated reasoning about others’ knowledge—the "I think about what others think" recursion—and negotiate equilibria that account for both adversarial and cooperative goals. The communication architectures that bound this strategic reasoning are the same ones that constrain task decomposition and role allocation in the preceding sections: the topology through which agents exchange information fundamentally shapes the depth of iterated reasoning that is feasible, while sparse but regular structures promote more effective coordination than dense graphs under uncertainty [35]. In mixed-motive settings, agents must simultaneously prioritize robust planning and coordination; emergent collective behavior under uncertainty often involves flexible delegation and turn-taking rather than rigid role assignments [82]. Adaptive frameworks allowing agents to update interactions based on response quality or involvement have demonstrably improved collaborative performance, echoing the dynamic role reconfiguration discussed earlier [83]. The robustness of recursive belief modeling against observational noise remains an open challenge; systems with rigid models of others exhibit brittle performance under stress, whereas flexible estimators and adaptive role-assignment demonstrate resilience [83]. Prompt design remains a pivotal, yet often overlooked, method for injecting strategic reasoning into agents—an intervention that operates at the procedural layer and can serve as an alternative or complement to architectural redesign [81].

Scaling social reasoning to multi-party settings with divergent utility functions introduces a

qualitatively different set of challenges: the need to leverage the varying expertise of multiple agents while handling heterogeneous goals. Systems observed in these environments often fail to effectively leverage expert agents even when explicitly identified, with self-organized teams repeatedly falling into the epistemic problem of averaging distinct strengths rather than appropriately weighting expertise [10]. This misplaced effort results in performance losses up to 41.1% in ML benchmark scenarios [10]—perhaps our understanding of the dynamics beyond sheer alignment. These behaviors can be mapped to communication patterns that characterize human-inspired interaction, where agents in open flexible setups show emergent coordination and role differentiation, yet clear limitations to tightly synchronizing toward a singular joint optimal plan [81]. In contrast, systems that enable agents to dynamically structure communication among high-contributing agents demonstrate strong performance gains, especially under low-resource conditions, relative to static baselines [83]. These self-organizing characteristics stand in direct contrast to centralized orchestration solutions, revealing that suboptimal delegation under uncertainty is not necessarily a failure of individual reasoning but an inherent property of rigid architectures. This observation carries forward directly to human-agent teams: if homogeneous agent collectives cannot flexibly weight expertise, human overseers embedded in these systems face analogous challenges when attempting to calibrate trust across heterogeneous participants.

The frontier of social cognition in LLM multi-agent systems lies in the tight coupling of recursive belief inference (deliberation) with adaptive, dynamic emergence of structured cooperation—a coupling that parallels, and in turn supports, the human-in-the-loop governance mechanisms of the following section. The current literature demonstrates clear capacity for social constructs (e.g. coordination, specialization) to arise without explicit direction, yet reliable generalization to structured conflicts—achieving fine-grained interactions—fails especially without first-principles Theory of Mind. Scale and composition complicate this: while increasing agent numbers magnify the potential for collective achievement, it simultaneously creates novel failure modes—the "integrative compromise" trap and cascade amplification—as observed in loosely coordinated, self-organizing settings [10, 84]. This scale-related fragility has a direct analogue in human-Agent teams: the same mechanisms that produce divergent and suboptimal outcomes in agent-only collectives will manifest in human-agent groups unless governance structures explicitly encode expertise weighting and exception handling, thereby extending the urgent need for human oversight discussed next. Future research should therefore prioritize: (i) building adaptive environment-specific ToM estimators that continuously update their model of others; (ii) designing architecture modifications that induce implicit ToM, rather than imposing rigid top-down interaction schemes; and (iii) developing formal theory linking the foundational degrees of strategic reasoning to large-scale system performance metrics—thus linking socially "aware" agents with robust, scalable cooperation under both diverse and human cohabitants.

3.5 Human Interaction, Collective Governance, and Process Transparency

As LLM-based multi-agent systems transition from research prototypes to deployment in consequential domains, the interaction between human stakeholders and agent collectives becomes a critical determinant of both effectiveness and legitimacy. This subsection examines the mechanisms by which humans participate in, govern, and achieve transparency within multi-agent decision processes, moving beyond simple majority voting to address the normative complexity of high-stakes environments.

Human-in-the-loop planning presents a fundamental design tension: users must be able to meaningfully shape collective behavior without being overwhelmed by interaction overhead. Co-planning interfaces operate along two principal axes: semantic modification (altering intent, objectives, or constraints) and structural modification (reorganizing task decomposition, role assignments, or coordination topology). Structural interventions, such as directly editing the task graph or conversation tree, provide more reliable control over group processes than semantic guidance, though they impose greater cognitive costs on users. A critical challenge lies in

determining the optimal locus and granularity of human intervention: aggregated oversight of orchestration can improve alignment but risks creating bottlenecks, whereas distributed control reduces accessibility but introduces potential coordination conflicts. The cognitive-load perspective suggests that human overseers face bounds analogous to those motivating multi-agent decomposition itself, arguing for transparent, well-scoped "exceptions" rather than continuous full-system monitoring [8, 48].

When a human collective—rather than a single operator—engages with agent systems, the problem shifts from individual control to collective governance through social-choice-theoretic aggregation. Research on multi-agent societies reveals that alignment of individual agents does not necessarily transfer to the group level, with behavior often socially driven rather than representing universally internalized values. This observation directly informs the design of aggregation methods: treating variance as a positive signal to recover in situations of loss of trust, rather than simply as weakness to be resolved, and thus situations requiring globally ethical outcomes in heterogeneous groups of stakeholders generically lead to less-satisfying and less-fair outcomes, and transparent decision procedures should expose these latent value differences rather than conceal them. Context-aware and participatory aggregation methods can mitigate these tensions by prioritizing stakeholders' varying preferences and returning the inherent conflicts to the surface.

The governance of complex interaction processes fundamentally shapes human-agent outcomes, and it extends beyond architectural choices to the governance of protocols, norms, and coordination measures, including the general axiomatic requirement (such as those studied in social-choice theory) of protocols for different groups. Research on MCP servers and protocols demonstrates that the interaction protocol can be made explicit and context-aware, therefore improving task coordination [85, 86]. The most pressing governance challenge may be accountability: when maximizing global objectives operates across distributed-stakeholder structures, mechanisms must preserve multi-stakeholder visibility into decision sources while respecting local data ownership and proprietary knowledge [33]. Accountability becomes partially hereditary yet bounded: a responsible principal has to designate boundaries. New procedural layers address this by explicitly encoding the person responsibility, context, and system ownership of agents for actions through structured regulation—with continuous compliance demanding ongoing rather than one-time behaviors. Unlike simple delegation, this "two-level" multi-agent accountability in trusted execution environments requires data retention, task-level policy, provenance, and action auditing to create collaborative controlled safety in such environments.

In the management of asynchronous information and shared memory, governance design is necessary: as a solution to the need for structured dynamic access control on private and shared memory between agents and humans, memory is not necessarily globally accessible [87]. By-pattern matching of who has access, and to what it is meant (privacy, security, or commercial viability), the absence of access to memory of all collaborators can be intentionally chosen.

Finally, the simulation of stakeholder structures emerges as a critical tool for policy-facing decisions. Multi-agent systems can emulate the multiplicity of stakeholders for decision-making and can even integrate adversarial viewpoints within decision processes [81]. While such simulation does not replace the formal democratic principles, it provides low-cost structured exploration for how dynamic trade-offs may unfold before committing to governance frameworks of possibly experimental rules.

Looking ahead, a notable gap remains between the development of principled human interaction techniques and their validation in full high-stakes settings, where institutional norms, escalation hierarchies, and judicial-like oversight will matter. A research agenda must integrate advanced computational social-choice theory, formal governance, statistical-reasoning, and incorporate human cognitive constraints, developing reliable and implementable structures for how humans can coexist, control, and be accountable for interactions within multi-agent systems that increasingly reason and act on their behalf.

4 Learning, Evolution, and Continuous Adaptation in Multi-Agent Systems

4.1 Supervised Fine-Tuning and Multi-Agent Reinforcement Learning

The parameter-updating paradigm for multi-agent systems has undergone a fundamental shift: from treating LLMs as fixed reasoning engines orchestrated through prompt engineering to explicitly co-training multiple agents through supervised and reinforcement learning objectives. This transition reflects a growing recognition that the innate collaborative capabilities of out-of-the-box LLMs are inherently limited; recent empirical evidence demonstrates that while multi-agent scaffolding can improve performance across benchmarks, collaborative behavior must be elicited explicitly rather than assumed [88, 89]. The central challenge formulation of simultaneous policy optimization in this context introduces unique problems that distinguish the domain from both single-agent fine-tuning and classical multi-agent systems research: the joint optimization of heterogeneous policies, the temporal credit assignment across asynchronous agent utterances, and the stabilization of training under non-stationary agent behavior.

Supervised fine-tuning provides a foundational yet essential approach for instilling collaborative competencies. Prior to the advent of RL-based paradigms, co-training strategies were built upon sequential cooperative frameworks. In the CORY framework, the LLM to be fine-tuned is duplicated into two autonomous agents (a pioneer and an observer) that generate responses concurrently, and the resulting agents are jointly trained with a shared reward, exchanging roles periodically to foster coevolution [90]. This alignment dynamic illuminates that the value of multi-agent RL lies not merely in optimizing a single policy against a fixed environment but in creating a natural generative curriculum in which agents pose increasingly complex challenges to one another. Similar logic embodies Multi-Agent Evolve, where three interacting agents (Proposer, Solver, Judge) self-co-evolve questions, attempt solutions, and evaluate outputs, effectively pushing RL as a mechanism for generating diverse, evolving training, without human-curated success criteria [91]. These approaches shape a crucial architectural insight: the self-play structure itself acts as a classifier, solving the scalability gaps for training in open-ended domains where verifiable outcomes are scarce.

Reinforcement learning extends these supervisory frameworks by optimizing for emergent interaction dynamics. However, the most direct transfer of single-agent RL methods to multi-agent settings is unstable. The standard GRPO (Generalized Reinforced Policy Optimization) grouping assumptions break down in MARL because prompts vary by role and turn, requiring development such as AT-GRPO (Agent- and Turn-wise Grouped RL), which provides both single-policy and multi-policy training regimes for robust deployment [92]. Prompt execution, in a multi-agent RL setup, requires the co-training of multiple LLMs together; recent evidence indicates that independent training is insufficient to induce effective collaboration, while multi-agent co-training with a RL objective can boost collaborative performance across benchmarks and hold out map directions [88]. This suggests, moreover, that the performance boost from RL does not merely result from classical, inter-agent message exchange but from the reward shaping and co-evolution of the agents with a controlled “process” to align outcomes [14].

A pivotal architectural distinction emerges between centralized orchestration and decentralized coordination in the RL-based MAS literature. In LLM-based MASs, the "Conductor" model introduces a centralized, trainable orchestrator (a 7B LLM) trained with RL to coordinate other, possibly frozen, LLM workers. The Conductor learns a communication topology, and agents can recursively select themselves as workers, improving performance via recursive topologies and dynamic test-time scaling [60]. In contrast, AgentNet explicitly promotes a decentralized, DAG-shaped organization where each agent locally decides its connectivity and tasks via retrieved expert knowledge, enhancing robustness and removing the single point of failure [11].

Reward modeling and credit assignment are the two fundamental problems framing the remaining difficulties of RL-based co-training. The reward function must align goal success

with coordination effectiveness along multiple dimensions—not only via task score but explicit communication readability and information grounding [14]. Moreover, recent frameworks integrate “structural optimization,” acknowledging that with challenge of multiple interacting agents, purely optimizing individual “local” policies is insufficient; alternating “local” prompt updates and “global” structure optimization is necessary [24]. By addressing the hidden non-stationarity in decentralized exchanges, this gradient-free, vector approach activates broader structure in the agent community [24].

A promising frontier is the **unification of structure optimization and RL-based adaptation within a dynamic self-improvement loop**. Model Swarms illustrates collaborative search in weight space, where pooled LLM experts collectively adapt and improve across tasks using only a handful of examples, bypassing traditional gradient-based retraining for domain-specific synthesis [93]. The scaling of the RL- co-training methodologies remains an underlying challenge: single-agent systems match MAS performance when the compute is used as a constrained thinking budget, demonstrating that the key contribution of RL in MAS cannot be simply increased computation but efficient creation and deployment [51]. Notably, results such as *AT-GRPO* achieving large accuracy gains on long-horizon planning (from 14.0% to 96.0-99.5%) indicate that structured RL can switch agents from computation-inefficient approaches (e.g., that assume consensus) to full utilization of the allocated inference [92].

Looking forward, a next agenda emerges, scaling **crucial system-level** components of RL-based MAS. The challenge of coordination—the non-stationarity in the training period—remains, yet we foresee it accelerating from purely static to learned multi-timescale orchestration. Optimistically, the generalization of cross-policy heterogeneous training offers a promising reason for further work; the field is well-placed to explore new horizons that will exceed current training pipelines and capture richer social and human-aligned collaboration paradigms.

4.2 Experience-Driven Self-Evolution through Reflection and Memory

The capacity for continuous improvement distinguishes multi-agent systems from static software pipelines, and experience-driven self-evolution—the ability to accumulate, consolidate, and reuse knowledge from interaction histories—has emerged as a central mechanism for achieving this adaptability. Building directly on the co-training paradigms established in the previous subsection, which emphasize the joint optimization of agent policies, this evolution occurs along two complementary axes: reflection, which enables the identification and correction of errors and inefficiencies, and memory, which provides the substrate for storing and retrieving refined knowledge and skills. As multi-agent systems engage in long-horizon tasks, the context-window limitations of individual LLMs necessitate external memory that persists beyond individual interactions, forming the basis for collective self-enhancement [20, 94]. Reflection mechanisms complement this memory by enabling agents to proactively manage their reasoning processes, monitor and evaluate their own and others’ outputs, and strategically steer future behavior [65]. Critically, llm-based agents often suffer from isolated operation and reliance on static databases, and struggle to grow from experience or share that growth collectively [95].

Reflection in multi-agent systems functions at multiple levels: from individual error identification to emergent system-level behavioral correction based on, for example, goal-state reflection, which grounds decisions in states and enforces ongoing goal alignment to counter the hallucinations and ungrounded reasoning steps common in long-horizon tasks [96]. At the collective level, however, collective reflection must address the challenge of multi-agent coordination, where structural adjustments can be optimized through search—echoing the orchestration-optimization lens that the following subsection treats as an explicit design space [63]. This dual rationale—fixing both low-level error identification and high-level coordination strategy—is a hallmark of self-evolving systems.

The architecture of memory, however, determines whether collective reflection results in sustained adaptation or merely short-term correction. Memory-augmented and non-parametric

learning systems constitute a significant class of memory mechanisms, enabling agents to adapt to novel circumstances without traditional gradient updates. From RL-based co-training, these mechanisms represent a form of "inference-time adaptation" that complements weight updates and cooperative policy optimization. The design-angular space of such systems is vast, encompassing both context-resident compression, where summaries and factual impressions are maintained within the agent’s context window, and retrieval-augmented stores, or memory banks, which persist information across episodes [94]. Within large-scale agentic systems, memory architecture often follows a multi-level organization based on the degree of semantic abstraction. Instead of relying on similarity-based search from flat vector stores, these hierarchical designs can embed a structured index, where each memory vector contains a positional encoding that points to related sub-memories in the next layer. During the reasoning phase, an index-based routing mechanism facilitates efficient, layer-by-layer retrieval without computing exhaustive global similarity scores [97].

Importantly, the placement of memory across the multi-agent ecosystem is as important as its internal structure—an insight directly parallel to the orchestration design choices discussed in the preceding subsection on RL-based training? Recent empirical work on multi-agent workflows has begun to decompose the design space of procedural memory beyond the individual agent. A systematic study dissects procedural memory into modules for orchestrator and task agents, revealing that memory placement is critical: orchestrator memory is crucial for effective task decomposition and delegation, while fine-grained agent memory improves execution accuracy. Teams composed of smaller reasoning engines can substantially benefit from such heuristic memory, closing the performance gap between weaker and stronger base models by leveraging prior execution traces for more accurate future planning and tool use [40]. This supports the view that shared external memory serves as a system-level substrate for coordination and knowledge consolidation.

The synergy between reflection and memory is most powerful when it supports self-improving practices, yet this synergy is not automatic. Experience-following behavior—where LLM agents tend to produce highly similar outputs to retrieved memories—underlines both the power and the risk of memory banks: while consistency is valuable, scope over-fitting to maladaptive past experiences can propagate errors [71]. This highlights the need for memory management to be an active, strategized process rather than passive storage. To address this, learnable memory frameworks have been proposed to model the full memory cycle—retrieval, aggregation, and storage—via data-driven methods. Examples include the use of mixture-of-experts gates to select relevant memories, learnable aggregation to improve memory utilization, and task-specific reflection to adapt storage. This allows for the agent to *learn how to memorize* as a function of optimal future performance [98]. More conceptually, the road from raw experiences to a more distilled and reusable abstraction has been unified in the framework of Storage → Reflection → Experience, forming a coherent trajectory from storage to refinement and rising to a higher “Experience” plane [99]. Within the correlational work on workflows, process-level optimizations can also be viewed as experiential; for example, the automatic search of agentic workflows based on modular design spaces can be considered a form of collective procedural experience that guides future workflow orchestration [63]. Strategic sense in such modular memory architectures is that functional units like memory and planning can be reused and adaptively designed for new tasks [100].

There remains, however, a crucial distinction between self-correction and true self-evolution: the former is episodic, while the latter is cumulative. A robust mechanism for consolidating insights from reflective loops into a persistent memory bank—often by abstracting reflections into reusable “lessons”—is fundamental to avoiding repeated errors and enabling systemic adaptation. For example, frameworks for writing “lessons” into memory can break cycles of repeated failure and produce improvements without changing the core model [70, 99]. This move from in-context learning to “in-infrastructure” learning marks a major shift: it enables agent experiences to

be distilled and internalized continuously, thereby connecting this subsection to the broader trajectory of the survey—which will later focus on dynamic orchestration redesign based on memories and continual learning loops.

Looking toward the future, the field of agentic memory is increasingly confronting the engineering realities of system deployment. Empirical analyses of memory systems—including benchmark attention, metric precision, LLM backbone sensitivity, and latency overhead—are leading to a deeper understanding of where memory can be most effective, highlighting trade-offs between accuracy and efficiency [101]. For large-scale multi-agent systems, the principle of continual learning cannot be decoupled from scalability; the attraction of persistent memory is that it provides a flexible framework to support tasks requiring adaptation, ease of use, and economies of coordination. Continual improvement of agentic environments requires calibration mechanisms that, for example, can leverage future task evaluations as quality labels to filter stored memory, managing the quality-versus-quantity trade-off for lasting utility [71].

From a coherent view, the evolution of experience in multi-agent systems is fundamentally a process of progressive abstraction: from in-context reflection to persistent, system-level memory, and ultimately to the dynamic orchestration choices that the following subsection addresses. Achieving this progression requires not only robust reflection and sophisticated memory retrieval, but also carefully designed architectures that place memory where it matters. As the field moves from stateless personal assistants towards genuine long-lived, adaptive agent networks, the interplay between reflection and memory—and autonomy—remains central. This directly sets the stage for the next subsection’s central thesis: that orchestration itself can be made adaptive, learning from cumulative experience to search over structures that were never manually designed.

4.3 Adaptive Orchestration of Configuration and Workflow Optimization

Static orchestration architectures—whether centralized, hierarchical, or decentralized—presuppose a fixed mapping between tasks, agents, and interaction protocols. Yet the distribution of problem instances encountered by deployed systems is neither stationary nor fully predictable, and the optimal configuration for one task family can be actively harmful for another. The imperative to move beyond static design has driven a rapidly maturing line of research that treats the orchestration itself—agent roles, communication structure, and workflow sequencing—as a searchable, optimizable object [14]. This subsection analyzes four complementary families of approaches: credit-aware prompt refinement, hierarchical configuration learning, recursive structure co-evolution, and metacognitive delegation.

A foundational challenge for any orchestration optimization is assigning credit: when a system output is poor, which agent’s prompt, which message in the communication chain, or which coordination decision was responsible? Early approaches addressed this through iterative, block-coordinate prompt refinement, where individual agent prompts are optimized in isolation while the rest of the system is evaluated holistically. However, this greedy isolation ignores interaction effects; a prompt that improves local performance can degrade collective outcomes by distorting communication patterns. Credit-aware variants have consequently emerged, explicitly attributing errors to interaction components—such as dependency edges—using structural analysis of the collaboration graph [102]. By identifying which edges carry amplification risk, gradient-free optimizers can be directed toward the highest-leverage configuration changes rather than diffusing updates across all agents.

A second family treats configuration as a hierarchical search problem. Rather than optimizing continuous prompt parameters, these approaches frame the design space as a set of discrete choices: which agent profiles to instantiate, how to topologically connect them, and what level of message sparsity to enforce. This includes learning to select among pre-defined interaction structures [14]. Work on communication topology has demonstrated that the design space can indeed be navigated if connectivity is optimized: moderately sparse structures can outperform

both fully dense and fully sparse graphs due to their ability to suppress error propagation while preserving beneficial information diffusion [35]. The incorporation of adaptive joint design is further supported by the observation that the profitability of learned pruning is highly dependent on contextual accuracy projections, linked to efficiently optimizing a network for resource use [22] removed]. The common thread is that token economies and error budgets force orchestration design into explicit, data-driven optimization, displacing the implicit token expenditure of static, fully connected architectures.

At the most radical end of the spectrum are generative models that synthesize not just weights, but the system’s architecture for the current task. Here, we see orchestration structures directly generated as stochastic functions of inputs—most notably to diffusion models that iteratively construct topology and of tasks themselves [36]. By framing orchestration as the result of an accurate using offline training signals [36], we get a path that bypasses manual engineering and treats the formal structural decision itself as a learnable object that orients itself toward cost, performance and robustness. Indeed, the conditional graph diffusion approach yields the system’s design as a prediction, embedding hard constraints preferences directly with the generation-process. The synthetic materials class extends the scale of brute force further, allowing designs search over areas too large for classic gradient-free search to handle.

Finally, a delegation-based line of research focuses on negotiation: an orchestration that struggles to fit all demands, but instead partitions the problem structure adaptively and assigns role-aware composition plans [103] and keeps a library of best when and where appropriate. Rather than optimizing a fixed, explicit architecture, these systems currently maintain a portfolio of reusable communication blocks; at query time, an organizer performs composition, bringing together primitives—such as review, voting and plan-and-execute—that have proven useful in similar contexts [30]. This generates both expressive and acquired working structures for interactions, significantly improving robustness for task segmentation. Crucially, agent primitives may be the most realistic step towards decoupling adaptation from outer-loop software design because they allow adaptive orchestration to remain as a lightweight memory whose deployment is deferred costless task time.

All families converge on a set of pressing findings. Benchmarks empirically show that underestimating orchestration diversity is costly; the token-reduction of mass across all dynamic, data-driven methods are actually yields performance gains rather than trade-offs in effectiveness [79]. At the same time, methodologies diverge in what they converge to: fixed pipelines with threshold filters are favorable for design convergence but perform poorly in robustness, whereas full generative closed-loop model remains vastly more powerful, but has heavier inference. Furthermore, in both ecosystems on this frontier, one must not use oracle, task-specific supervision explicitly injecting meta-cognition into the search cost. The most principled path forward uses the same primitives to record and deploy human feedback, allowing recurrent orchestration to be viewed as problem in itself and adaptive.

Thus, the survey here posits a clean division of labor: the separations of prompt-level credit assignment, topology-level design, and model-level adaptation have reached a point where the proposed framework clearly delineates guidance, control, and analysis. But we must now perform their modular integration. Future research must study seamless interplay from mining how diffusion over weights dictates structure, and how structure provides decision evidence for prompting. Devising these richer semantics around orchestration is an exciting directive; from a pragmatic standpoint, the same active, data-driven structuring will ensure operational margins are real, not the exclusive domain of bespoke tasks, by studying variation across heterogeneous deployed workloads.

4.4 Continual and Lifelong Learning Across Agent Lifecycles

The operational lifespan of LLM-based multi-agent systems introduces a fundamentally distinct learning paradigm from the offline training regimes and static orchestration designs discussed in

prior subsections: continual and lifelong learning across agent lifecycles. While the preceding subsection examined how orchestration architectures can be optimized as searchable objects, deployed systems face an equally critical challenge: these configurations must remain effective over time as task distributions evolve, user preferences shift, and experience accumulates without catastrophic forgetting. This challenge is compounded by multi-agent-specific complexities, including dynamic membership (agents joining or leaving populations), the need for online adaptation under cost constraints, and the maintenance of coherent collective knowledge across distributed memory architectures [39]. Critically, the static orchestration of prior subsections assumes a fixed mapping between tasks and agents; this subsection addresses what happens when that mapping itself must be continuously revised.

A primary axis of this learning is the consolidation of procedural experience, which directly extends the memory-infrastructure argument introduced earlier regarding experience accumulation. Systems increasingly adopt modular reference, decomposing past task trajectories into reusable memory units distributable across orchestrators and task agents [63]. LEGOMem demonstrates the design space: orchestrator memory proves critical for effective task decomposition and delegation, while fine-grained agent memory improves execution accuracy, narrowing the performance difference between smaller and larger models by leveraging prior traces [63]. This suggests a hierarchical division of experiential labor, where high-level planning and low-level execution benefit from distinct, decoupled memory horizons—a principle that mirrors the layer-by-layer retrieval logic proposed in hierarchical memory systems. The architectural framing of multi-agent memory—distinguishing shared and distributed paradigms and proposing layers akin to I/O, cache, and memory—further identifies "cache sharing" and "structured memory access control" as critical, unsolved protocol gaps for reliable lifelong operation [39], revealing that the memory infrastructure itself must anticipate continual learning.

Parallel to memory consolidation, continual adaptation necessitates the very hierarchical configuration learning that the previous subsection identified: not as a one-time search, but as an ongoing, data-driven process. Frameworks like AutoAgents couple task-to-role relationships dynamically, generating agents based on task content and observer roles to improve plans and responses, demonstrating that a fixed population is suboptimal for long-horizon adaptation [44]. This is essentially hierarchical configuration search applied at runtime rather than design time. The orchestration control is itself a learning target; the puppeteer-style paradigm trains an orchestrator via reinforcement learning to adaptively disposition and prioritize agents, reducing computational costs while improving performance, with emergent "more compact, cyclic reasoning structures" appearing over time [9]. These findings imply a need to treat adaptive orchestration as a first-class objective alongside individual-agent learning—the self-reflective loop that the previous subsection named as a future challenge is in fact taking shape.

However, this continual adaptation introduces an essential tension: continual learning improves performance for tasks seen before but risks over-fitting to stale distributions—the very failure mode the prior subsection warned against. While prior work on orchestration optimization primarily aligned error propagation, here the risk is that the topology-evolution mechanisms themselves amplify cascading and drift: continuous reconfiguration may amplify errors that could have been contained under static, but bounded, orchestrations. In this context, frameworks like AgentNet preserve knowledge while a centralized course of design is decentralized: agent roles are continuously updated on performance scores and dynamic connectivity, allowing agents to specialize and evolve in a decentralized fashion [29]. A stochastic self-organization approach frames the optimization of communication structure as an integrated task, updating a directed acyclic graph (DAG) based on agent Shapley-value approximations during interactions, enabling the learning of effective collaborator networks without further parameter training [83]. These topology-evolution approaches, however, must be embedded within a broader robustness model that tolerates the very error cascades such updates can inadvertently amplify—hence a lifecycle-aware socio-technical monitoring architecture with process-level failure attribution is

necessary [43, 16]. This constraint is precisely the counterweight to the generative orchestration flexibility discussed in the previous subsection that called for embedding robustness.

Most existing lifelong learning research, however, is limited to memory-resident, within-system scale. As deployments reach enterprise and cross-organizational scales, a formalization of "continual learning" must integrate memory management and the curation of shared knowledge. Agent-associated components and versions must move beyond a fixed set, and continuous deployment requires persistent, versioned state for retrieval and reproducibility [23]. This enables the "continuously updated modeling" of subscribing agents to the "living" knowledge recorded in the system [23], mirroring the recursive structure co-evolution perspective from the previous subsection: designs should be continuously informed by their own performance. This "multi-scale memory" case remains under-addressed, and system scale-up has already shown that memory allocation carries significant performance consequences, for instance, organizations can leverage execution history to infer globally useful abstractions rather than instant problem-solving specifics [63]. This creates a "scale of recollection" tension—an agent collects personalized low-level traces to generalize high-level processes, but at the expense of memory updates.

Looking toward future work and the following subsection, three promising directions emerge that bridge the insights of memory consolidation and orchestration design: (i) the design of "procedural memory" capable of avoiding catastrophic forgetting within long deployments, likely with context-of-execution; (ii) the formalization of collective skill consolidation, where routing and strategic composition of individual learnt tasks lead to broader capabilities; and (iii) the integration (in which memory-management and topology-adaptation into a unified continual learning framework) where dynamic and reflexive orchestration feedback loops can guard against collective drift and distribution shift [43, 16]. Addressing these will require moving beyond static evaluation to long-horizon, interactive benchmarking protocols under development [84, 104, 23]. In this way, the continual learning paradigm completes the transition we have traced from fixed orchestration, to search space optimization, and ultimately to a self-reflective system ability—where orchestration and memory mutually inform each other, making system configuration a question answered by accumulated experience rather than reprocessed from scratch.

5 Evaluation, Benchmarks, and Analytical Methodologies

5.1 Quantitative Benchmarks for Task-Specific Multi-Agent Evaluation

The quantitative evaluation of LLM-based multi-agent systems has undergone a fundamental shift from single-agent capability assessment toward task-specific benchmarks that explicitly capture the structural and dynamical properties of collective intelligence. This evolution reflects a growing recognition that the value proposition of multi-agent systems—division of cognitive labor, expertise specialization, and emergent coordination—cannot be adequately measured through traditional single-turn, single-model evaluation protocols. As such, the benchmark landscape has diversified across domains including software engineering, mathematical reasoning, scientific discovery, embodied decision-making, and multimodal interaction, with each domain imposing distinct coordination requirements and evaluation criteria.

Domain-specific benchmark suites constitute the empirical backbone of multi-agent evaluation. In software engineering, benchmarks such as SWE-bench provide realistic, repository-scale coding challenges that demand task decomposition, cross-file understanding, and iterative verification—properties that naturally stress test multi-agent workflows. Similarly, mathematical reasoning benchmarks (e.g., GSM8K, AIME) serve as controlled environments for studying solution diversity and aggregation strategies, while embodied and planning benchmarks expose the temporal and real-time constraints of agent coordination [57]. Notably, the comparative analysis in [13] demonstrates that multi-agent performance is fundamentally architecture-dependent: across 260 configurations spanning six benchmarks and five canonical architectures, the relative

performance gain of multi-agent systems compared to a single-agent baseline ranged from +80.8% on decomposable financial reasoning tasks to −70.0% on sequential planning tasks, revealing that collaborative advantage is contingent on task decomposability and architectural fit. This empirical finding underscores a critical methodological principle: benchmarks must be instantiated with structured task dimensionality that permits the independent scaling of task difficulty, coordination complexity, and inter-agent dependencies to accurately estimate capability boundaries.

An equally pressing issue is benchmark design methodology itself. Static benchmarks suffer from saturation, distributional narrowness, and vulnerability to data contamination—problems exacerbated in multi-agent settings where the interaction space expands combinatorially. Dynamic benchmark generation, particularly through LLM-in-the-loop pipelines with verifiable ground truth, has emerged as a promising direction for continuous benchmark evolution. In addition, the role of heterogeneity in multi-agent evaluation has been highlighted by [19], which introduced a comprehensive testbed spanning 5 domains, 21 tasks, and 5 agentic functions, and evaluated 27 LLMs through over 1.7 million assessments. Their work demonstrates that heterogeneous configurations can yield up to 47% performance improvement on the AIME dataset over homogeneous settings, pointing to the importance of structural interactions between different capabilities in benchmarks that are sensitive to heterogeneous collective compositions. Moreover, the design space of such benchmarks must account for both collaboration, where success is achieved through information sharing and delegation, and competition, where adversarial incentive structures alter agent interaction trajectories; benchmarks that conflate these modes risk invalid conclusions about system competence [57].

Metrics for task success and efficiency form the third pillar of this subsection. While coarse success rate remains the almost universal outcome measure for its ease of reporting, multi-agent systems demand more granular and economically grounded metrics. Primary outcome metrics such as pass@k must be augmented by cost-accuracy trade-offs (e.g., token consumption and financial cost), latency measurements, and task-level accountability. The results of multi-agent measurement research show that costs are often significant and can be reduced without quality loss: up to a 52.07% reduction in overhead was achieved through LLM routing of multi-agent systems compared to state-of-the-art methods [25]. Efficiency metrics are also increasingly coupled with correctness measures, since token consumption and latency are critical indicators for scalability.

Looking forward, the next generation of benchmarks must contend with several paradigm-shifting challenges. First, adaptive and self-evolutionary properties among agents, architecturally related to LLM machinery, are insufficiently covered by current benchmarks, which mostly assess static, point-in-time performance. Second, benchmarks are built from observed failure patterns [7], with a dataset of over 1,600 annotated traces across 7 MAS frameworks revealing 14 failure modes, including design-level flaws, inter-agent misalignment, and verification gaps—providing reference points for benchmark coverage evaluation. There is an inherent tension between benchmark stability and domain transferability: effective aggregation and evaluation in dynamic and realistic enterprise settings remains an open issue.

Call for the next frontier of benchmarks: benchmarks that are difficult to saturate, cost-controlled, and difficulty-scalable, which produce realistic and dynamic tasks and encode verifiable ground truth, allowing continued model co-evolution, and a systematic isolation of the sources of effective cooperation versus the performance of the best individual member [10].

5.2 Interaction-Centric and Process-Level Performance Metrics

While task-specific benchmarks in Section 5.1 capture the *outcome* of collaboration, a comprehensive evaluation of LLM-based multi-agent systems (MAS) demands a rigorous examination of the *process* by which outcomes are achieved. The behavioral dynamics of agent societies—the flow of information, the quality of reasoning embedded in exchanges, and the structural contribution

of each participant—are often as consequential as the final result, particularly for diagnosing bottlenecks and systematically attributing value to individual components. This subsection moves beyond final accuracy to introduce process-centric metrics and analytical frameworks that interrogate the collaborative trajectory itself. We focus on three methodological pillars: communication efficiency and semantic diversity, process-level structural diagnostics, and formal credit assignment—each providing a distinct lens on the mechanisms that separate successful from failing collaborations.

Communication efficiency and semantic diversity. A primary indicator of healthy collective cognition is the informational uniqueness of agent exchanges. Formally, a system with high *semantic entropy* ensures that messages contribute novel evidence rather than redundantly reiterating the same reasoning chains. Methodologies can model agent interactions in multi-round exchanges as graphs, enabling detection of semantic variation, redundant reasoning paths, and off-topic degradation [79]. This "experience-following" property shows that high input-task similarity between a new query and retrieved memories can trigger nearly identical agent outputs, causing *error propagation* and *misaligned experience replay* [71]. Without careful regulation of information as it traverses communicative pathways, a system suffers from context rot and network degradation. The severity of these costs is task-dependent: coordination costs only degrade quality under high cognitive load, whereas for low-intrinsic-load tasks, coordination overhead dominates and dims returns [8]. Such diagnostics reveal critical bottlenecks where not only performance but also semantic information richness degrades.

Process-level collaboration and structural diagnostics. Analyzing collaboration trajectories exposes structural bottlenecks and efficiency patterns that outcome metrics obscure. Multi-agent interaction patterns can be studied via interaction-aware architecture searches—frameworks such as MaAS treat workflow orchestration as a distribution over architectural patterns, selecting query-dependent system configurations to optimize the trade-off between contribution value and resource consumption [46]. On the macro scale, the *scaling* behavior—in either num agents or model variety—has to distinguish task-centric uncertainty from deterministic channel limitations. The *effective channel count*, K^* , provides a principled performance bound for collaborative systems, approximating the number of independent evidence sources accessed. This bound suggests that homogeneous agents saturate performance early, validating well-documented diminishing returns, whereas heterogeneous agents providing complementary evidence support more robust scaling [58]. Capturing these dynamics aids in stage-wise diagnosis: where does miscommunication occur? Where does redundancy dominate? Such structural diagnostics are precisely the basis for the longitudinal and adversarial evaluations explored in Section 5.3.

Game-theoretic credit and contribution. Effective process analysis cannot stop at describing interaction patterns; it must trace causal attribution to the components that contribute to system outcomes—particularly in settings where contributions are asymmetric or emergent. Deployment of *post-hoc attribution* based on interaction graphs reveals which channels most strongly relate to task success, guiding targeted optimization of the overall system. This attribution perspective can uncover coordination crediting problems, where agents central in communication nevertheless fail to receive recognition for pivotal inputs. From these analyses emerge model-dependent reliability assessments: a robust intelligent system must maintain *resilience* to different model backbones, demonstrate stable *effective collaboration* through coordination, and converge with manageable token consumption. Evaluating the process smoothes the path toward the causal insights and emergent dynamics discussed next in Section 5.3, since attribution forms a necessary bridge from performance metrics to the underlying mechanism.

5.3 Evaluation of Emergent Collective Dynamics and Social Behaviors

The evaluation of emergent collective dynamics in LLM-based multi-agent systems (LLM-MAS) represents a fundamental departure from conventional performance assessment, shifting the analytical lens from predetermined task completion toward the characterization of unanticipated

social behaviors that arise from sustained inter-agent interaction. These phenomena—including the formation of social hierarchies, the spontaneous emergence of role specialization, the propagation of information through communication networks, and the degradation of coordination under adversarial conditions—constitute the core observable manifestations of collective intelligence in agent societies. A methodological challenge persists: these behaviors are frequently invisible to aggregate accuracy metrics yet profoundly influence system-level outcomes, necessitating a suite of analytical tools grounded in complex network science, social simulation theory, and causal inference [89, 77].

A robust framework for assessing social-emergent structures must integrate multi-level observations: detecting structural patterns in interaction topologies, diagnosing behavioral dynamics that underlie these structural formations, and tracing causal pathways from micro-level agent characteristics to macro-level outcomes. For the first dimension, researchers have adopted quantitative techniques from complex network analysis—including graph centrality measures, community detection algorithms, and topological analyses—to identify and quantify structural configurations such as core-periphery layouts, leadership emergence, and clustering in agent populations, measuring role specialization as a deviation from an undifferentiated homogeneous baseline [35, 34]. Research underscores that sparse communication topologies produce distinctive error-suppressing properties that prove instrumental in containing misinformation and preventing error propagation, yet overly sparse structures compromise information diffusion, thereby throttling collective capability [35]. For the second dimension, computational methods such as temporal graph abstraction, entropic measures of interaction dynamics, and information-theoretic diagnostics have been applied to probe dynamic phenomena—ranging from information cascades and systemic false consensus to the spread of errors [102, 105]. The GUARDIAN framework exemplifies this by modeling the collaboration process as a discrete-time temporal attributed graph, and explicit error-propagation dynamics create a temporal basis for hallucination amplification and coordination collapse detection [105]. In parallel, metrics such as conversational robustness indicators—derived from cluster entropy, lexical repetition, and semantic similarity—quantify communication system’s language use under game-theoretic pressures, revealing functionally distinct communication patterns in cooperative versus competitive settings [106].

These structural and dynamic indicators are insufficient without a causal micro-to-macro linkage. A key research paradigm lies in the study of controllable perturbations: exposing collectives to systematically varied traits—such as divergent reward structures, memory strategies, or performance limits—while measuring the effects on macro-level properties. This approach allows for scientific explanations of emergent dynamics through causal interpretation, distinguishing between cases where knowledge coordination produces beneficial outcomes versus how mispecified reward systems produce unintended collusion. Recent work demonstrates that training incentives can elicit sophisticated linguistic steganography in emergent communication, a robustly harmful behavior that largely persists even with countermeasures [107]. This example illustrates the necessity of explicit, targeted measurement frameworks for emergent, safety-relevant collective behaviors. This methodology, combined with framework-level observation, characterizes self-organizing and self-protecting behavior in heterogeneous agent populations [108, 14]. Concrete causal analyses reveal that even a single, small injection error can exert disproportionate harmful effects in multi-stage workflows; stronger foundation models do not uniformly yield improved robustness, and composability of agents in adaptive environments was itself a decisive factor in systemic outcomes [102, 82]. Evidence from agent interaction modeling further suggests that the dynamics of emergent behavior propagate through communication structures, and individual identity loss—referred to as echoing—can create unintended malformed communication structures, illustrating that properties of agent populations rather than agent-level task performance must take center stage in any design process [109].

Moving forward, the evaluation of emergent collective dynamics should shift toward longitudinal audits and standardized interaction protocols, to support the validity of results across model

backbones and system designs [110]. It is essential to standardize reporting and statistical benchmarking across diverse tasks [111]. We must realize that maturing emergent behavior into sustained resilience is not just an obligation—it offers a novel route to designing proactively robust agent-based systems whose reliability is anchored in a principled account of the interplay between local interactions and global dynamics [104].

5.4 Benchmarking Frameworks, Standardization, and Reproducibility

The evaluation of LLM-based multi-agent systems faces a fundamental reproducibility crisis that stems from the extraordinary sensitivity of these systems to seemingly minor implementation choices. Unlike single-agent evaluations, where a model’s output can often be attributed to its parameters, MAS evaluation results are entangled with framework-level design decisions that can induce order-of-magnitude differences in performance metrics [112]. This realization has motivated both the benchmarks discussed in the previous subsection and the emergence of standardized benchmarking platforms and unified codebases—tools designed to isolate the effects of architectural choices from the underlying LLM capabilities, and to provide the controlled, generalizable evidence on which systematic benchmarking depends.

The central methodological challenge is that architectural topology—the structure of inter-agent communication—confounds result interpretation from the specific model used, making basic insights impossible without standardized execution pipelines and carefully controlled experiments. MAFBench directly addresses this by integrating multiple existing task benchmarks into a framework-agnostic pipeline, providing a standardized execution environment. This controlled comparison reveals that structural choices alone can increase latency by over $100\times$, reduce planning accuracy by up to 30%, and lower coordination success from above 90% to below 30% [112]. Consensus on the issue of comparability further leads to the introduction of a taxonomy of fundamental differences between frameworks, another key step toward an understanding of the causal mechanisms through which design choices affect performance [112].

Alongside the need for controlled comparisons, the reproducibility crisis in MAS research is exacerbated by the practical absence of a common codebase. MASLab represents a community-level effort to address this by consolidating over 20 established methods across multiple domains into a unified environment, ensuring both systematic verification against official implementations and standardized evaluation protocols [47]. Using such uniform codebases enables the comprehensive controlled comparisons of 8 models on over 10 benchmarks, enabling researchers to obtain a clear landscape of the field without the burden of reproduction and redundant implementation efforts. MAESTRO also contributes to this vision by supporting the integration of 12 representative MAS via lightweight adapters, exporting framework-agnostic execution traces alongside critical system-level signals such as latency, cost, and failure rates [48]. Through controlled experiments with repeated runs and varied backend models, MAESTRO reveals that MAS executions are often structurally stable but temporally variable, contributing to substantial run-to-run performance variance—this finding is critical for the practice of aggregating results without principled reliability analysis [48].

A related issue is the transferability of performance rankings across distributions. Traditional evaluation paradigms, which rely on static benchmarks, risk rankings that fail to generalize to new task distributions, a concern that directly affects the causal generalization goals raised in the previous subsection [113, 74, 48]. Recent work shows that diminishing returns of architectural selections often depend on the sophistication of the underlying model family, with relative architecture preferences remaining consistent even on unseen frontier models [13]. Such systematic studies of scaling behavior provide a predictive framework to estimate performance based on measurable decisions and system constraints, allowing researchers to identify the best architecture for 87% of held-out configurations [13]. This suggests a shift toward predictive models that estimate rankings via principled extrapolation.

Addressing the control of confounds further demands robust experimental practice. Differences

in base LLM versions across models, stochastic decoding, and the high variance between runs can all skew results [112]. To address this, robust practices have emerged: conducting sufficiently many repeated runs, applying paired statistical tests, and reporting both variants and uncertainty characterization. Engaging the role of intermediate verifiers and controlled interventions extends reliability analysis from static metrics to interaction-level fault tolerance, and in a way consistent with the causal micro-to-macro outlook introduced earlier, dynamic fault injection probes system behavior across the full stack: profiling fault-tolerant behavior and diagnosing exactly when certain model capabilities benefit or harm system robustness [102, 82].

A promising trend in benchmarking is the move toward modularized, continuous, and transferable evaluation. Guided by the demands of evolving system designs and the practical constraints of the next subsection, evaluation frameworks should incorporate systematic control and dynamically resurface benchmarks that maintain relevance and avoid test-set saturation [113]. However, a crucial tension remains: the trade-off between standardization and agility. On the one hand, standardization is necessary for comparability; on the other, benchmarks must remain sufficiently challenging and agile to capture rapid advancements in a continuously shifting field [74]. The future of evaluation thus lies in the co-design of unified codebases, dynamic benchmark suites, and adversarial testing that promotes reflect on and iteratively improve evaluation methodology itself—a foundation on which both the causal analyses described earlier and the economic considerations to follow can be reliably built.

5.5 Efficiency-Aware and Cost-Economic Evaluation

As LLM-based multi-agent systems transition from research prototypes to production deployments, economic viability emerges as a decisive determinant of practical adoptability. Powerful multi-agent pipelines often exhibit prohibitive resource consumption, with communication and coordination overhead frequently dominating direct inference costs. This subsection introduces systematic evaluation methodologies that place resource consumption, latency, token economics, and efficiency-accuracy trade-offs at the center of system assessment, moving beyond task-accuracy-only evaluations constrained purely by quality metrics.

A foundational observation is that the costs induced by multi-agent orchestration are not intrinsic to collaboration but are heavily design-dependent. Redundancy in message passing and communication graph size often contributes waste without proportional quality gains. Empirical analyses [79, 35, 114] show that pruning redundant communication in the spatial-temporal message-passing graph significantly improves bottlenecks, achieving performance comparable to state-of-the-art topologies at roughly **13%** of their cost. Evidence from controlled evaluations [48] further indicates that multi-agent architecture is the dominant driver of resource profiles and cost-latency-accuracy trade-offs, often outweighing backend model choice, underscoring the need to treat orchestration as a first-class efficiency optimization target. This finding positions architecture-driven efficiency as a lever on par with—or exceeding—model-level optimizations, particularly as LLM performance converges [115].

Recent advances propose formal, dynamic mechanisms for economic control, responding to these observations. For instance, EvoRoute applies a self-evolving, experience-driven paradigm to dynamically select Pareto-optimal LLM backbones and routing strategies at each step, reducing execution cost by up to 80% and latency by over 70% while maintaining or enhancing system performance [53]. In a complementary approach, MasRouter formulates Multi-Agent System Routing (MASR) as a unified optimization that jointly navigates collaboration modes, agent roles, and model routing, reducing overhead by up to 52% as a cost-effective and plug-and-play solution [25]. These approaches demonstrate that efficiency evaluation is not merely about measuring a static configuration; it involves designing for and measuring controlled dynamic orchestration under explicit economic constraints.

To assess these elements, specific economic-centric metrics and auditing frameworks have emerged. From a communication-economics perspective, token consumption and end-to-end

latency are fundamental currencies [23, 116, 47]. For example, in enterprise deployments, coordination and routing overheads—such as payload referencing and message processing—may significantly impact operational latency [78]. Metrics that capture token cost, monetary expense, and computational resources are critical for systematic comparison, with performance evaluations feature explicit cost budgets and Pareto trade-offs. MAESTRO directly integrates control signals such as latency, cost, and failure rate into a unified evaluation suite, demonstrating that temporal and structural stability can vary substantially on repeated runs, making run-to-run variance a key performance metric [48]. The importance of infrastructure is also highlighted: operational frameworks like AgentRM provide structural, scheduling, and resource governance, using Multi-Level Feedback Queue lane schedulers and rate-limit-aware admission control to reduce P95 latency by 86%, increase throughput by 168%, and eliminate zombie processes and resource leaks [117]. In many cases, such operational improvements can be of the same order of magnitude.

Formal frameworks for attribution and contribution analysis are critical for justifying resource deployment. Symbolic attribution approaches abstractify the compute-to-communication trade-off. Any single agent’s decision process can be viewed as a graph where each edge, node, and token becomes a cost layer, and pruning communication graphs or model routing based on attribution and ablation allows for the identification of individual decisions [79, 52]. MasRouter, for example, learns to "respond or route" to an intended design [25]. This high-level analysis extends to the measurement of the Pareto frontier. In many cases, a moderately sparse multi-agent architecture with token efficiency can achieve up to a 90% of the fixed budget while reducing cost by up to 80%, a finding directly evaluated when measuring token economics and viable system.

Finally, broader implications for simulation tests and scientific design emerge. Evidence indicates that prior accuracy gains of multi-agent systems often stem from unaccounted computation and excessive thinking budgets rather than inherent architectural benefits. When controlling for reasoning-token budget, single-LLM systems systematically match or outperform standard multi-agent methods, demonstrating centralized **information-embedding and context efficiency** in collective behavior [51]. Efficiency-aware evaluation also addresses cognitive load and information processing; The Cognitive Load Theory (CLT) analogy provides a principled lens for interpreting the boundary in which multi-agent coordination helps. It demonstrates that gains from multi-agent collaboration concentrate on reasoning-heavy tasks with high intrinsic load, while coordination overhead dominates on low-intrinsic-load tasks—showing an explicit "cognitive-load-profile" efficiency frontier [8]. Matrix-based studies that quantify the efficiency of contributed feedback loops demonstrate that trajectory reduction methods (e.g., AgentDiet) can reduce the input token budget by over 40% without compromising reasoning accuracy, largely by eliminating redundant or expired information during execution [118]. Such trajectory-level optimization is critical for evaluation because it identifies and distinguishes waste from necessary computation, introducing measurable behavioral and communication efficiency metrics.

In conclusion, a principled economic and efficiency characterization of multi-agent systems requires understanding not only algorithmic accuracy but also system topology, scheduling policies, and token economy. Future should standardize budgets, report costs along Pareto-optimal fronts, and embed adaptive and economic controls directly into multi-agent design via closed-loop feedback [48]. These formalizations will enable researchers to distinguish between efficiency gains that result from coordinated intelligence and those that result from increased computational expenditure for the same, accelerating progress toward **economically viable and practically deployable** multi-agent systems. Research should also further quantify energy, allocation, and hardware-level constraints, bringing full-scale resource economics into the multi-agent design framework.

6 Security, Safety, Alignment, and Ethical Considerations

6.1 Attack Surfaces and Threat Modeling in Multi-Agent Architectures

The transition from monolithic LLM deployments to interconnected multi-agent architectures fundamentally redefines the security perimeter. Whereas single-agent systems present a concentrated attack surface limited to the model’s prompt and tool interfaces, multi-agent systems expand this surface across communication channels, shared memory infrastructures, and emergent topological relationships that have no single-agent analog. A systematic threat model must therefore distinguish between inherited vulnerabilities—those that persist from the underlying LLMs—and genuinely novel attack vectors that arise exclusively from inter-agent composition [104].

Communication channels constitute the most consequential attack surface in multi-agent architectures. The reliance on natural language as the substrate for inter-agent coordination renders the system vulnerable to prompt injection that propagates through the trust gradient between peer agents. Critically, prior work has demonstrated that many models that exhibit resistance to direct attacker-controlled prompts remain susceptible to compromise through what has been termed "trust exploitation": the malicious content is delivered not by an adversarial actor directly, but by a peer agent that has itself been compromised, effectively laundering the attack through the system’s internal authentication of membership [104, 77]. The notion that an agent faithfully executing its role can become an equivalent of an automated malicious payload delivery mechanism highlights a fundamental deficit in inter-agent security boundaries: messages are implicitly granted execution privileges by virtue of their provenance alone, a property without direct precedent in software interfaces where provenance does not imply integrity. Within a single orchestration system, this trust boundary is eroded further by structural delegation: multiple agents operating under a single orchestrator pose the dual challenge of preventing compromise of the orchestrator while simultaneously preventing a malicious agent from extracting information across the system [16].

The addition of shared memory and knowledge repositories to a multi-agent architecture creates a second, remarkably distinct attack surface. Unlike communication channels, which are transient and sequenced, shared memory systems function as persistent, globally accessible blind spots where persistence extends the attacker’s reach well beyond execution; attacks on a single knowledge object can simultaneously poison future decisions by all agents reading the corrupted context. The systemic risk is no longer constrained to the current run, but persists when the next run begins, converting memory poisoning into a probabilistic and determined backdoor. Tool surfaces extend this sabotage, with adversarial or manipulated responses from external APIs inadvertently injecting false ground truth into long-lived decision-making loops. Empirical evidence from embodied and vision-language multi-agent systems confirms the breadth of this threat, demonstrating that agents are susceptible with average success rates up to 84.30% across three simultaneous vulnerability classes: prompt injection, memory poisoning, and tool attacks—showing that the aggregate risk across multiple components in an interconnected system is not merely additive but composes into a distributed reliability hazard [104].

The enumeration of the enumeration of attack vectors further exposes the topology as a variable and exploitable structure. Centralized designs, while simpler to interconnect, create a concentration of trust that is a stepping stone for an attacker’s decisive blow; decentralized designs distribute trust but admit widespread propagation. These dynamics are not hypothetical: "Telephone Loop" attacks exploit cross-agent delegation cycles in multi-agent architectures to degrade system performance while being inert against single-agent architectures, and the delegation re-links essential to the design become attack vectors. The inter-dependence induced by the task decomposition itself—serving to complete others’ assignments—becomes an attack graph of functional dependencies, where any node’s compromise propagates within the likelihood matrix. Formal threat models must therefore model two quantities: the network intrusion

capacity, i.e., the extent to which attacker perturbations can cross non-compromised agents, and the structural trust from the attacker to any target—critical voiding high levels. Multi-agent system security ultimately reduces to the degree of overlap between the functional and trust graphs. Two distinct but subtle attack paradigms operate within these structures: hidden intentions and collusion, in which peer agents, individually or collectively, pursue goals that oppose the system-level objective without revealing their intent. These "intention-hiding" attacks are particularly insidious precisely because they can reflect legitimate system objectives in the observable trace, and the complexity of action attribution in long-horizon interconnected systems makes their identification fundamental [6, 16].

The design of threat models for multi-agent architectures must therefore abandon single-agent homogeneous views in which the attacker interacts from the outside, and instead, interior attacker models explain suspicious behavior at each node while their objective is to compromise the entire organization. The framework can not only identify but anticipate the design components that are especially at risk, including systems with deep centralized orchestration and overlapping tool dependencies; conversely, from the defense’s perspective, systems that minimize the intersection between interaction and trust graphs have an inherent "danger level" that is greater than any arbitrary message. The research frontier, however, does not stop at the point of discovery but continues from its urgency: As yet, we can identify attacks; we still lack a comprehensive defensive and robust architecture that the design of tasks and topologies with security metrics as first-class constraints. Future work must therefore pursue both directions: further categories of attacks that exploit the specific mechanics of multi-agent communication and collaboration, and, more important, the collaborative, cross-agent security architectures of mutual verification and behavioral baseline detection. In this, the key principle remains clear: MAS security has moved from securing the single model to securing the relations and structures that connect them. This "network topology for security" governs the transfer: though performance depends on inter-item cooperation, security is only as strong as the most confident edge.

6.2 Cascade Propagation, Systemic Failures, and Defense Mechanisms

The distributed nature of LLM-based multi-agent systems introduces a distinct class of systemic fragility: failures that originate locally—whether as hallucinations, reasoning errors, or targeted malicious injections—can propagate through inter-agent communication channels and amplify into organization-level collapse [119, 67]. While the preceding subsection established how inter-agent topologies define the security perimeter and the conditions under which compromises spread, this subsection deepens that analysis by focusing specifically on the *dynamics* of failure propagation and the structural and defensive mechanisms that determine whether such disturbances remain contained or escalate catastrophically. Understanding these propagation mechanics also provides the foundation for the normative dimension discussed in the following subsection, where misalignment, like error, is shown to diffuse along the same communication pathways.

The mechanisms of error propagation in multi-agent systems exhibit two contrasting patterns. On one hand, empirical analyses reveal that even benign errors—such as hallucinations—compound as they traverse agent chains, leading to a phenomenon known as *agent drift*: a progressive degradation of behavioral coherence, decision quality, and inter-agent alignment over extended interactions (Agent Drift: Quantifying Behavioral Degradation in Multi-Agent LLM Systems Over Extended Interactions). This drift manifests as semantic deviation from original intent, breakdown in consensus mechanisms, and the emergence of unintended strategies. On the other hand, iterative verification pipelines can attenuate errors when explicitly structured revision mechanisms are in place (e.g., generator-evaluator or critic architectures), intercepting low-confidence outputs and yielding net error reduction. This asymmetry is crucial: the mere existence of cascades is not deterministic—whether amplification or suppression dominates is a design choice under direct architectural control (Agent Primitives: Reusable Latent Building

Blocks for Multi-Agent Systems; A Survey of Frontiers in LLM Reasoning: Inference Scaling, Learning to Reason, and Agentic Systems).

The structural topology of the system plays a determining role in these propagation dynamics, operating as the same interplay between *functional graphs* and *trust boundaries* highlighted in the security analysis. Decentralized architectures without centralized verification tend to propagate errors more readily than those with centralized coordination, as demonstrated through controlled scalability studies (Towards a Science of Scaling Agent Systems; Understanding Agent Scaling in LLM-Based Multi-Agent Systems via Diversity). Compounding this, specific attack vectors exploit iterative delegation cycles—undetectable at the single-agent level—to convert the computational graph into a cascade-amplifying loop (Agent Security Bench (ASB): Formalizing and Benchmarking Attacks and Defenses in LLM-based Agents). In memory-shared architectures, the persistent and globally accessible nature of the memory surface (already identified as a core attack surface in the preceding subsection) intensifies the risk: poisoning a single memory object propagates corrupted context across all subsequent agent reasoning steps, with single-step attack success rates empirically reaching 84.30% (A-MemGuard: A Proactive Defense Framework for LLM-Based Agent Memory; Agent Security Bench (ASB): Formalizing and Benchmarking Attacks and Defenses in LLM-based Agents).

A critical challenge in mitigating cascades is distributed fault attribution—the inability to locate the responsible agent within a multi-step execution trace—which constitutes the key obstacle to correction. Recent advances in multi-agent reinforcement learning address this through turn-level credit assignment and agent-specific advantage estimation, enabling groups to identify the source of deviation and learn transferable robustness against perturbation (MARSHAL: Incentivizing Multi-Agent Reasoning via Self-Play with Strategic LLMs; M³RL Mind-aware Multi-agent Management Reinforcement Learning; A Survey of Frontiers in LLM Reasoning: Inference Scaling, Learning to Reason, and Agentic Systems).

Building upon these diagnostic capabilities, defense frameworks aim to intervene directly in propagation pathways. At the interface level, isolation and trust boundaries separate user inputs, tools, inter-agent messages, and system-level prompts, mitigating prompt-injection and memory-poisoning campaigns (Agent Security Bench (ASB): Formalizing and Benchmarking Attacks and Defenses in LLM-based Agents). At the reasoning level, the A-MemGuard framework exemplifies consensus-based validation: agents independently retrieve and cross-check memories, requiring alignment before any action, achieving 95%+ reduction in attack success against memory poisoning at minimal utility cost by distilling detected failures into retained lessons (A-MemGuard: A Proactive Defense Framework for LLM-Based Agent Memory). At the topology level—consistently with the security principle of reducing overlap between functional and trust graphs—introducing redundancy and diversity into the agent pool breaks percolation channels: heterogeneous configurations (diverse models or prompts) provide substantially higher resilience than homogeneous scaling, consistent with information-theoretic bounds showing that multi-agent error amplification is limited by the number of independent information channels (K^*) rather than the raw agent count (Information).

The field is now undergoing a transition from reactive post-hoc detection to *proactive resilience engineering*. Rather than merely defending against known attacks, work is consolidating toward systemic design-time risk management, continuous multi-agent adversarial evaluation via benchmarks (ASB), and self-healing architectures that preempt distribution from percolating further through causal-driven online monitoring (Agent Security Bench (ASB): Formalizing and Benchmarking Attacks and Defenses in LLM-based Agents). Establishing systemic resilience, therefore, requires not only attack-resistance at individual nodes but *structural* endurance of the entire communication topology—the emergent property that governs whether the failure of individual agents remains a local event or becomes an organizational one. This structural dependence aligns directly with the neighboring topics of this survey: the inter-agent security bounds which define immediate pathways, and the normative boundaries which depend on

the same cascading dynamics. The challenge is thus the following: to make agentic systems *structurally* resilient, not merely attack-resistant, so that the very design of collaboration embeds safety as an intrinsic property of the system, not an ex-post addition.

6.3 Multi-Agent Alignment and Value Governance

The alignment of multi-agent systems with human values constitutes a fundamental departure from single-agent alignment, as the ethical status of a collective cannot be inferred from the sum of its parts. This subsection analyzes the mechanisms of emergent misalignment, frameworks for multi-value governance, and the structural conditions that determine whether an agent society remains tractable to normative control.

Central to this analysis is the finding that individually well-aligned agents do not guarantee collectively aligned behavior. Agent behavior is contextually driven and socially induced—agents alter their expressed values to conform to interacting peers, undermining any notion of stable internalized norms [120]. Worse, emergent power-seeking behavior arises spontaneously in agent societies without explicit incentives, driven by the capabilities enabled by interaction [109]. Empirical evaluations confirm this disconnect: when alignment is measured at the group level rather than individually, behaviors that appear practical are frequently socially influenced rather than reflecting authentic motivation, yielding no reliable mapping between individual training objectives and collective outcomes [99]. These findings demand a paradigm shift from verifying whether each agent is aligned to verifying whether the interaction topology itself suppresses or amplifies misalignment.

This systems-level challenge is compounded by a result from social choice theory: requiring globally ethical behavior in heterogeneous groups leads to generally unsatisfying and unfair outcomes when aggregated naively. Simple voting or majority rules cannot properly unite conflicting moral frameworks [121]. A multi-agent alignment framework draws on social science to represent the heterogeneity of normative systems and uses multi-objective evolutionary optimization to search for norm parameterizations that achieve simultaneous accommodation of competing agent interests [110]. However, this approach is vulnerable to failure modes that arise from noise across objectives when optimizing for multiple values at once. A promising direction is the application of value decomposition networks—but the massive space of norms and interactions—whether learned centrally or emerged through negotiation—remains an open problem for computation. The design of aggregation functions that are fair, robust, and computationally tractable is a core challenge for collective governance [33].

Beyond normative alignment, structural governance must contend with the dynamics of power and private resources in systems of autonomously acting entities. Power asymmetries among agents have been shown to directly compromise collective sustainability: a small subset of agents can optimize local goals at the expense of global ones to imperil the entire system [78]. Control frameworks inspired by principal-agent theory, in which an overseer must maximize global objectives against individual priorities, provide a formal model that reveals what is likely "reward hacking"—where a single allocation for one agent effectively corrupts the decision space of its peers [120]. Accountability in these systems is structurally complicated by principal-agency multiplicity: in a one-principal-multi-agent framework, distorted responsibility leads to accountability that is both exercised but uneven [122].

Two concurrent approaches to advance these goals involve general detection and theoretical integration. The first should be detection of whether an agent subverts the group’s objectives: recent methods use “agent critics”—cross-validated models that, for example, identify questionable reasoning in other language models—which have been extended to detect misalignment at the system level rather than the agent level [38, 109]. The second is the use of formal “logic and architectural guard-rails” to ensure that an agent’s declared values are consistent with its actions across contexts [Agent Security Bench (ASB): Formalizing and Benchmarking Attacks and Defenses in LLM-based Agents. Notably, both these approaches observe that alignment defense

is a system property that depends on social parameters—e.g., number of agents, communication density, and reward to the environment. If these parameters are tuned to adversarial regimes, the effect of alignment modifications (observations that, e.g., fine-tuning a single base model for values of each agent in a network does not necessarily alter group behavior under certain conditions) can be canceled entirely.

In summary, value governance in LLM multi-agent systems is fundamentally tied to the topology of social influence. Research is moving in the direction of incorporating methods from evolutionary multi-objective optimization, social choice theory, and agent-based control. Future work will entail (i) deriving a measure for reliability under sociolinguistic pressure to decouple an agent’s declared values from its actual utility function, (ii) making system-level misalignment detectable via reflection on parameter adaptation, and (iii) achieving internationalization and preservation of power asymmetry in independent algorithmic governance. Achieving high-level alignment, therefore, is not a matter of tuning individual models—it is an emergent problem that must be tackled with governance regimes and social structures that are both the target and the tool in.

6.4 Explainability, Accountability, and Human Oversight

The interpretability of LLM-based multi-agent systems presents a fundamentally different challenge from that of their monolithic counterparts: understanding a system requires not only comprehending individual agent capabilities but also tracing how information is transformed, combined, and potentially corrupted across distributed interaction paths. While single-agent interpretability focuses on model internals, multi-agent interpretability must address a higher-order problem of *relational attribution*—determining which agent, which message, and which decision point contributed to a system-level outcome. This distinction carries profound implications for accountability, as the causal chain connecting a final output to its origins often traverses multiple layers of delegation, revision, and synthesis that are opaque to external observers. The preceding discussion of value governance established that misalignment can propagate through interaction topologies, and this subsection examines the complementary requirement: making such topologies legible. Without interpretability, neither ethical drift nor cascade failure can be traced to its source—critically, the failure attribution methods explored here provide the empirical backbone for the systemic explanations that alignment interventions demand, just as the following discussion of privacy will require the same relational tracing to govern data flows across trust boundaries.

System-level interpretability requires the capacity to provide faithful, coherent accounts of *why* a collaborative outcome was reached, revealing the ways in which inter-agent dynamics shape final decisions. It is the mechanism by which the systemic misalignment described earlier becomes observable, and indeed preventable, since outcomes must be legible before their ethical valence can be assessed. Work on multi-agent collaboration mechanics reveals that behaviors such as consensus-seeking and compromise emerge through interactions, and that these dynamics can systematically alter outcomes—for instance, self-organizing teams frequently exhibit a tendency toward integrative averaging rather than expertise maximization, dropping performance relative to the best individual member by up to 41.1% [10]. Means must be provided to make these dynamics visible to the user. Process-level tracing—logging who said what to whom, when, and with what strategic effect—is the basic and minimal element of such interpretability, and many frameworks now include execution traces that track message routing and intermediate state transitions as standard infrastructure. Yet raw trajectories are insufficient without *interaction-level* abstractions identifying patterns of influence: e.g., the directed dependency graph, as described in work on error cascades that models collaboration to give an early-warning risk of amplification [102]. The generalization of these state-representations does not capture how aggregated opinion emerges from individual alignments; hence, designing an interface for *system-level* explanation remains a core unresolved problem. Explanation must capture *design intent*

(expected workflow) as well as *actual behavior* (interaction realized), letting humans contrast them and understand when and why they diverge—a contrast that directly mirrors the alignment gap between declared values and enacted behavior in multi-agent systems.

Agency-level attribution—the reverse operation, tracing back from a system-level outcome to individual contributions—presents a distinct and equally critical difficulty. This challenge is tightly coupled to the structural conditions of alignment discussed earlier: just as misalignment emerges from topologies that no single agent controls, so too does accountability fragment across the same topology. In large, cooperating systems, *blame attribution* and *root cause analysis* are bottleneck tasks [123]. This is because "root causes" may be distributed across multiple agents to a degree that no single one is sufficient to explain the system's behavior or failure. For a majority of users to trace failures, a hierarchical causal graph is rarely useful in itself. One notable approach transforms a flat trace into a causal graph and then counterfactually evaluates four parts to isolate true root causes, pruning the search space with virtual oracle backtracking; consistent gains over flat-log approaches are confirmed on agent- and step-level attribution benchmarks [124]. In fault injection and reliability evaluation, systematic coverage may be difficult to uncover: historically linear workflows tended to collapse from a 15-taxonomy fault set, whereas iterative, closed-loop designs neutralized over 40% of faults, extending explainability toward predictive, fault-tolerant, and controlled systems [82]. Such findings support an approach towards building multi-agent architectures that are explainable not only by their internal trace structure but also by their design, an approach that potentially offers more meaningful accountability than *post hoc* causal attribution hacks.

Human oversight mechanisms are the final practical layer connecting these analytical capabilities to real-world operational control. A crucial form is human-in-the-loop verification, employed especially when validating or re-examining every agent interaction is prohibitive or even impossible; at the agent level, interpretable monitors and automated failure feedback become critical [45]. Their studies show that explicit orchestration signals—such as *verify* and *stop*—can significantly affect behavior, giving humans a *controllable* mechanism to regulate *when* agent action may proceed. This capacity to steer directly closes the loop with the alignment requirement for controlling agent societies. In many real systems, however, there is an aggregate orchestrator that acts with respect to the entire system, with human operators entering at a high level, not at every interaction. While this setup diverges from the per-interaction verification common in academic prototypes, the oversight of an entire multi-agent ecosystem requires both interpretability and auditability at the population level.

Authentication & Auditability thus needs a formal form that is not limited to the postmortem of a failure. Audit trails are essential for both post hoc failure analysis and continuous monitoring. Even systems that are designed as "black boxes" must provide causal trails—identifying which processor computed an intermediate result and which input was responsible. Procedural memory frameworks—for example, decomposing past trajectories into discrete memory blocks—can enable real-time reproduction of an exact path across sessions, allowing an eventual audit of prior output/output relations by replaying the same path [40]. This idea of modular, replayable trajectories to deliberately reexamine behavior is a step forward, though still far from a validated auditing of *why* an agent made the choice it did. Through the next generation of multi-agent systems, we will very likely require (a) **explanations as first-class objects**: language-based coordination protocols that attach human-interpretable, justificatory metadata to messages, generated to be auditable, not solely through opaque interpretational layers; (b) **formalized attribution**: game-theoretic or causal contribution frameworks that take into account the aggregative dynamics of the system, potentially applying concepts such as Shapley values to distribute credit fairly in complex operations and collaborative decision-making, thus providing the modeling basis needed for explanation and user trust.

The above represents a meticulous articulation of what constitutes **multi-agent explainability, accountability and human oversight** in the larger scheme. From where we stand, the area

is still an *architecture-defined* discipline—control structures determine the range of possible explanatory data in a multi-agent system, and when that range is stable, so is the downside. Many important questions await—how to adjudicate transparent decision-making mechanisms in partially decentralized systems, and how to provide robust explanations of system failures in shared multi-organizational settings with external confidentiality constraints. Without a measure of *measurable* perturbation and experimentation, LLM-based multi-agent systems risk remaining opaque systems with convenient objects. Human oversight is the minimal regulation to guarantee reliability both in normal and failure; rather, it serves as one of the steadiest baselines for steering a class of systems that is only becoming more autonomous: the measure of a system’s true interpretability is, and should be, the counterpart of the trust we are ready to place in it—a trust that the prior discussion of misalignment and the following discussion of privacy both demand.

6.5 Privacy Preservation in Interconnected Agent Environments

The proliferation of LLM-based multi-agent systems introduces a privacy landscape fundamentally distinct from that of monolithic models, where data flows are no longer confined to a single trust boundary but traverse inter-agent communication channels, shared memory infrastructures, and heterogeneous organizational domains [33]. The core challenge is not simply the protection of data from external adversaries, but the governance of data access *within* the collective itself: each agent must reveal only what is necessary for task completion while safeguarding information irrelevant to the interaction or restricted by its originating principal [121]. This shifts the privacy problem from a perimeter-defense model to an internal, granular enforcement challenge.

A foundational issue is defining the privacy threat model in these interconnected environments. Privacy does not equate to universal data hiding; rather, it demands a dynamic policy where each request or action may permissibly reveal private data to specific agents necessary for the task at hand [87]. This context-dependent access is further complicated by the presence of multiple principals with distinct directives and authority hierarchies. A critical threat emerges when a lower-authority principal instructs the system to extract information intended for another, or when an agent leaks data inadvertently to a peer within an ongoing collaboration [121]. This "multi-principal" security problem underscores that strong *secure* communication (e.g., encrypted channels) does not guarantee *safe* information use; policies must be enforced at the system level, not just the channel level [87].

To address these threats, a taxonomy of safeguarding techniques is emerging. **Frameworks for structured, policy-enforced sharing** represent a foundational approach. For instance, *Collaborative Memory* introduces private and shared memory tiers with granular read/write policies, with asymmetric, time-evolving user-agent-resource access controls encoded as bipartite graphs [87]. This design supports provable adherence to policies and full auditability, ensuring that memory fragments are only accessible if the current user-agent-resource configuration permits [87]. At the architectural level, **decentralized, retrieval-augmented** designs like AgentNet minimize centralized data exchange to avoid creating aggregation points for sensitive information, enhancing privacy-preserving collaboration across organizations [11]. This is complemented by **communication-minimizing strategies** that reduce the data footprint exposed; redundancy-aware pruning, such as AgentPrune, removes unneeded inter-agent messages, while trajectory reduction mechanisms compress context [79, 118]. Beyond architecture, **interaction protocols can be trained to optimize for privacy**, with multi-agent reinforcement learning teaching agents to minimize token generation and share only high-level task-relevant information, thereby limiting the granularity of sensitive data exposure [54]. Unifying these is the need for **cryptographically grounded enforcement**. While homomorphic encryption and secure multi-party computation are often cited, limits stack to practical LLM-based systems, requiring trusted execution environments and certification mechanisms that establish rules for verifiably validating data access [121]. Critically, LLMs handle "negative" security evidence (e.g., "do

not share without consent") better than "positive" evidence ("share if permitted"), suggesting that agents used for authorization must be primed towards a default-deny mental model with evidence of task need.

The path forward for privacy-aware agent ecosystems must prioritize solutions that integrate architectural isolation at the *operating system* level—akin to process and context lifecycle management [50, 11]—with policy-controlled memory [87] and challenge-driven routing to optimize for accuracy and cost while confining data to authorized peers [25, 42]. Technical mitigations alone, however, are insufficient; privacy also entails a governance and accountability dimension. This includes system-level auditing that balances access controls with robustness to malicious agents [48]. Implementation-level cues—such as: (1) supporting privacy-preserving interoperability; (2) capturing the interplay between performance and privacy, together with structured interfaces and verifiable data; and (3) integrating cryptographic tools for controlled and auditable collaboration on oversight tasks, and (4) recognizing the legal and jurisdictional aspects implicitly for privacy [33]—also need to be addressed. The field must move beyond coarse-grained all-or-nothing protection models, instead evolving the agent to reason about the actor environment to naturally calibrate its behavior, and pursuing a future where privacy is not a final product but an intentional, dynamic, context-aware property of the architecture and its interactions.

7 Applications, Societal Implications, and Future Directions

7.1 Domain Applications and Deployment Patterns

The deployment landscape of LLM-based multi-agent systems has matured from proof-of-concept demonstrations to domain-specific vertical applications, yet the empirical evidence reveals a nuanced picture of where multi-agent architectures deliver measurable value over single-agent baselines—and where the overhead of coordination becomes counterproductive. Across technical domains such as software engineering, scientific discovery, and mathematics, multi-agent systems have demonstrated their most compelling advantages through role specialization and structured verification pipelines. In code generation and debugging, for instance, frameworks decomposing tasks across analyst, coder, and tester roles exploit complementary error-detection capabilities that outperform monolithic generation [14, 6]. Similarly, in medical consultation, consensus aggregation mechanisms coupled with residual discussion structures achieve accuracies of 90.1% on MedQA and 83.9% on PubMedQA, with additional gains from accumulating correct-answer and chain-of-thought knowledge bases that directly address the performance degradation observed in excessively long dialogue histories [125]. These examples illustrate a crucial pattern: the multi-agent advantage is most pronounced when tasks require heterogeneous perspectives and when verification of intermediate outputs—through cross-agent critique or knowledge accumulation—mitigates the well-documented phenomenon of error propagation in long-horizon reasoning [14, 6].

However, the translation of these benefits to embodied and interaction-rich domains—robotics, multi-robot systems, and real-time decision-making—reveals domain-specific constraints that fundamentally alter the design calculus. Systems integrating LLMs into multi-robot systems inherit the challenges of latency, hallucination, and limited mathematical reasoning, while also contending with real-time perception-action loops and sparse feedback that render coordination overhead prohibitive [126, 127]. Deployments inspired by swarm intelligence, such as ant colony foraging and bird flocking, have shown that LLM-driven prompt-based behaviors can substitute hard-coded programs to induce emergent self-organization [59]. However, these successes are bounded by the environment’s ability to provide unambiguous, unmediated feedback—a stark contrast to purely text-based domains where the cost of iterative agent negotiation is comparatively minor. The decision-making frameworks developed for warehouse automation,

logistics, and transportation systems highlight a parallel tension; whereas classical multi-agent coordination in these domains has traditionally relied on reinforcement learning or hierarchical control, LLM-integrated approaches introduce a structured negotiation layer that incorporates reasoning and commonsense knowledge, at the expense of higher inference costs and unpredictable runtime behavior [4].

Beyond task-specific considerations, cross-domain deployments expose a systemic "effectiveness gap" between architectural promise and empirical outcome. Controlled scaling studies conducted across 260 configurations spanning 6 benchmarks, 5 canonical architectures, and 3 model families report relative performance changes from a +80.8% gain on decomposable financial reasoning tasks to a −70.0% regression on sequential planning tasks when adding centralized communication or verification [13]. This demonstrates that architecture-task alignment is the dominant predictor of collaborative value, not the number of deployed agents. Moreover, coordinated failures propagate in ways that single-agent ecosystems do not mask; fault-injection studies using 15 fault types confirm that the nature of failure is architecture-influenced, with linear pipelines suffering catastrophic collapse while closed-loop and iterative topologies neutralize over 40% of such faults [82]. These findings reveal that stronger foundation models do not uniformly improve robustness, and system-level reliability emerges as a critical design requirement [82].

The deployment landscape is equally marked by a degree of pragmatism between distributed flexibility and computational economy. Adaptive routing implementations such as MasRouter integrate collaboration-mode selection, role allocation, and model selection through a cascaded controller, which achieved accuracy gains of 1.8% to 8.2% while reducing overhead by 17.21% to 28.17% via customized routing [25]. Similarly, query-dependent sampling of an "agentic supernet"—a probabilistic distribution over system architectures—enables dynamic allocation of inference resources, requiring only 6–45% of the inference cost typically demanded by hand-crafted or fixed multi-agent configurable settings while outperforming them by up to 11.82% [46]. Equally important, cascading between single-agent and multi-agent systems can improve accuracy by 1.1–12% while reducing deployment costs by up to 20% [12]. These results collectively underscore that production-grade systems optimize, at least as much as agent design, an orchestration mechanism that matches resources to task difficulty and capability profiles.

For the near future, several pressing directions define the deployment frontier. First, a more rigorous application of classic Multi-Agent System theory is needed to close the gap between the loosely coupled LLM-driven interactions and the foundational principles of autonomy, environment design, and structured coordination protocols [89]. Second, there is a need for a comprehensive evaluation to assess the effects of system architecture and diversity on task performance [13, 57]. Finally, exploiting system heterogeneity (across different LLMs) has been shown to enable significant performance gains, with varied model profiles improving results by up to 47% on challenging datasets [19]. As the field matures, the central engineering insight is empirical: optimizing where each model is placed within a system designed to the epistemic structure and cost constraints of the domain—rather than adding agents for the sake of multiplicity—determines viability.

7.2 Simulation of Human and Social Dynamics with LLM Agents

The emergence of LLM-driven generative agents has inaugurated a new epoch in computational social science, positioning multi-agent systems as virtual laboratories capable of probing the micro-foundations of complex societal phenomena. This paradigm moves beyond traditional equation-based or purely statistical models by enabling the direct instantiation of heterogeneous cognitive entities whose internal deliberations, communications, and behavioral adaptations are executed in natural language [26]. A principal methodological advantage lies in the unprecedented granularity of inspection: these systems allow for the direct manipulation and observation of individual-level mechanisms—beliefs, memory states, and reasoning trajectories—thereby offering causal narratives for macro-level outcomes such as polarization, information cascades,

and institutional evolution. This makes them particularly well-suited for exploring the complex coupling between cognitive diversity and emergent system properties, which is often obscured in aggregate statistical approaches [67].

However, the construction of reliable social simulations requires navigating an expansive design space that fundamentally shapes emergent outcomes, much as the preceding discussion highlighted architecture-task alignment as the dominant predictor of value in applied deployments. This space can be characterized through a framework involving the inference trajectory (choices of agent architecture, cardinality, and connectivity), agent dependencies (the selection of base LLMs, system-message scaffolding, and persona definitions), and the environmental feedback loops that connect agent-to-agent and agent-context interactions. Different configurations can yield quantitatively divergent results, motivating research into automated design optimization [46, 24]. Within this context, a more subtle view suggests that scale—in terms of agent count—can be a source of diminishing returns unless it is coupled with heterogeneity in agents’ capabilities, skills, or strategies, a view that emphasizes the need to treat diversity as a distinct design axis in its own right [58]. This mirrors the cross-domain finding that architectural decisions, not the sheer number of agents, drive collaborative value.

Beyond design optimization, the long-term viability of this field rests on the ability to endow systems with theory of mind and to tackle the uncertainty intrinsic to human society. The inference of the internal beliefs of co-agents becomes crucial to reproducing their behavior, especially in settings that produce power imbalances or mixed incentives [67]. By guiding emergent processes with incentives—whether adversarial or cooperative—simulations can begin to replicate complex societal behaviors and the pathways toward cooperation or breakdown [67]. This demonstrates that governance and coordination structures are not static topologies but emergent outcomes of a deeper socio-cognitive substrate, one that is only beginning to be opened up by generative agents. Therefore, translating these insights into social design demands an interdisciplinary, highly cautious, theory-driven approach. While concrete results in this direction are still emerging, the combination of advanced reasoning, memory management, and reinforced self-play suggests a trajectory toward robust, generative social science—beyond mere scientific curiosity. Yet, given the potential for misuse at scale, developing such multi-agent systems must be subject to rigorous verification, mirroring the reliability engineering calls in real-world deployments [128].

Finally, a critical examination of how these systems’ underlying components behave—memory, overview, and coordination—can help ground the behaviors of multi-agent simulations in ways that do not rely on arbitrary rules [129]. For instance, discrepancies in memory can introduce considerable error into these systems, suggesting that memory design constitutes a core systems choice for simulation validity and performance. Thus, a healthy research agenda must couple high-agency designs with principled evaluation protocols that account for coordination—not merely content—when interpreting simulation outputs. Similarly, metrics that only judge an agent’s performance on static datasets need to be supplemented with measures of inter-agent variability and environmental robustness, as observed in work on composition and system robustness [97]. Only through such structured norms and evaluation practices can social simulations become a reliable, sound tool for social explanation—equally rigorous and socially responsible.

The trajectory of LLM-based multi-agent systems for social simulation thus aligns with the broader maturation of the field: from feasibility demonstrations toward principled engineering and validated methodologies. Just as the deployment subfield emphasized production viability and rigorous failure analysis, the simulation frontier must similarly evolve from proof-of-concept renders to instruments whose outputs can be interpreted, challenged, and trusted. By advancing the design space, embracing heterogeneity, and enforcing evaluation rigor, generative agent simulations promise to yield a new class of actionable knowledge about the social world—knowledge that remains both scientifically defensible and ethically mindful.

7.3 Human-Agent Collaboration, Interfaces, and Team Dynamics

The integration of human oversight into LLM-based multi-agent systems represents a fundamental shift from autonomous optimization to sociotechnical co-design, where the central research challenge is no longer merely maximizing collective intelligence but calibrating the boundary between machine autonomy and human control. This subsection synthesizes the current state of human-agent teaming (HAT), examining theoretical frameworks for co-planning, the dynamics of trust and role allocation, and the empirical methods needed to validate mixed human-agent systems. Unlike single-agent interaction paradigms where humans interact directly with one model, the introduction of multiple interacting agents creates emergent behaviors, distributed accountability, and coordination overhead that fundamentally alter the human operator’s cognitive load and intervention strategies [130].

A central contribution of recent work has been the articulation of design spaces for human-machine co-planning, cataloging workflows across a spectrum of human-in-the-loop patterns. These range from proactive supervision, where humans maintain continuous control, to passive, late-stage oversight, with interventions taking either semantic forms (e.g., textual feedback) or structural forms (e.g., modifying task graphs) [130]. Effective co-planning systems emphasize transparency and calibrated control while minimizing cognitive overhead, acknowledging that the effort required to direct multi-agent interactions must not exceed its operational benefit [130]. A crucial insight is that human coordination with LLM agents cannot be treated as a monolithic issue; the human must potentially interact with heterogeneous agents or a central orchestrator, with the choice between these architectures determined by the task at hand and the desired balance of control and efficiency [130]. This practical consideration has led to the recognition that the question is not merely whether a task can be automated, but whether the autonomy and oversight architecture delivers sufficiently greater value than both single-agent and fully manual alternatives—a term that directly guides strategic choices in enterprise deployment [131].

Central to the HAT agenda is the intricate interplay of trust, authority, security, and capability. The "Human-Agent Teaming" perspective frames the deployment not as employing a tool, but as onboarding a new cognitive entity, relying on the human’s ability to model the competencies, limitations, and distinct behaviors of individual agents. This is complicated by a fundamental "reliability gap": LLM agents generate outputs that appear fluent but are often factually brittle when using tools in dynamic environments [132], and technological agents frequently manifest spurious, unsafe behavior in novel environments, requiring dynamic trust calibration that is broader than static confidence scores [133]. Therefore, establishing role clarity and authority boundaries through explicit accountability structures becomes a cornerstone for achieving organizational performance without over-delegating to an unsafe degree, especially when operational failures arise from the larger network rather than a single model [128]. Notably, empirical validation shows that current autonomous agents—even with defense mechanisms, prompting strategies, and memory functions—have such low safety scores that human-in-the-loop verification is not just desirable but a necessary condition for acceptable risk in production environments [133].

Evaluating these mixed teams demands a suite of personalized metrics that measure success and safety. Measurement moves beyond task completion to include human satisfaction, interaction costs, and the quality of human intervention—for instance, distinguishing strategic early oversight from reactive, late-stage correction [130]. Process-oriented metrics (e.g., failure rates, resource usage) are prerequisite for understanding where agents fail, where human oversight is most valuable, and how to structure authority to minimize effort. This need for high dependence on correctness demonstrates why the security of the agent itself and its interactions with the human overseer must be treated as a co-design problem [134]. The most pressing challenge is the transition from research prototypes to robust cognitive platforms, a path that requires advances in human-agent teaming to mitigate the dangers and failures that plague current systems [110, 78]. Beyond single-operator oversight, we look toward the emergence of full-fledged

mixed-team settings and the need for domain-specific frameworks that require human oversight via clear interfaces, trust management, and mechanism design to ensure safe and beneficial devolution of authority in the division of human and artificial cognitive labor.

7.4 The Path to Production and Grand Challenges

The transition of LLM-based multi-agent systems from research prototypes to dependable production infrastructure confronts a fundamental empirical gulf: performance does not scale monotonically with agent count or architectural complexity. Systematic evaluations across 260 configurations and six benchmarks reveal that adding agents often yields diminishing returns or substantial regression due to coordination overhead, with relative performance changes ranging from +80.8% on decomposable financial reasoning to -70.0% on sequential planning [13]. This non-monotonicity is further compounded by architectural choices: framework-level design decisions alone can increase latency by over 100x, reduce planning accuracy by up to 30%, and lower coordination success from above 90% to below 30% [112]. The implication is stark—architecture selection is not an implementation detail but a first-order determinant of system viability, potentially outweighing even the choice of underlying models.

The failure-mode landscape of production-bound multi-agent systems compounds this empirical fragility. Beyond conventional single-agent error modes, these systems exhibit cascading reliability failures rooted in coordination overhead and error propagation. A propagation dynamics model abstracting collaboration as a directed dependency graph reveals three vulnerability classes—cascade amplification, topological sensitivity, and consensus inertia—demonstrating that a single injected atomic error seed can escalate into system-wide collapse across six mainstream frameworks [102]. Fault injection studies on three representative architectures further taxonomize failure modes into intra-agent cognitive errors and inter-agent coordination failures, revealing that stronger foundation models do not uniformly improve robustness [82]. Structural diagnostics confirm this vulnerability: self-organizing teams consistently fail to leverage expert contributions, with performance losses up to 41.1% on ML benchmarks arising from behavior that averages expert and non-expert views rather than appropriately weighting expertise [10]. Compounding this, skill selection in single-agent systems exhibits a phase-transition behavior, accuracy remaining stable only up to a critical library size before dropping sharply—a bounded capacity reminiscent of human decision-making constraints [73].

Compounding these reliability concerns is the absence of precise validation mechanisms. LLM-based multi-agent systems lack transactional safeguards and constraint-ensuring execution semantics in their default architecture [41]. Existing solutions often rely on self-validation by ungrounded agents, leading to unreliable self-correction and context loss. Even where validation frameworks exist, fault diagnosis is confounded by a mismatch between the flat-log data of current systems and the causal structure of the underlying failure modes; transforming chaotic trajectories into structured causal graphs with hierarchical oracle-guided backtracking is markedly effective, yet remains an open, under-deployed approach requiring substantial engineering [124]. Orchestration-level verification—plan-execute-verify-replan loops with LLM-based verifiers—can moderately improve completeness (from 3.1 to 4.2 on a 1-5 scale) and source quality (from 2.6 to 4.1), but a widespread adoption of such verification-driven mechanisms in current production pipelines remains sparse [135].

A third major challenge is effective resource management at several levels, where the economics of multi-agent systems are fundamentally distinct from their single-agent counterparts: operational costs include not merely per-model inference but also communication token overhead, scheduling latency, and coordination parameters. Communication topology choices emerge as the dominant global modifier; purposeful structured sparsity can yield a 50-70% reduction in inference cost without task degradation [35], while dynamic agent routing and cost-aware allocation can reduce inference overhead by up to 52% in deployed settings [25]. Conversely, naively scaling agent count to improve information processing overlooks contextual integration

failure, as large-scale multi-agent systems often experience coordination degradation [136]. Cost-aware operation remains an open problem: utility-based orchestration and composite model selection frameworks—both balancing answer quality against cost—provide compelling evidence toward viable, cost-monitored operation [45, 137]. Without these higher-level cost control, viable operation under financial or latency-constrained deployment remains difficult.

A fourth, conceptually deeper bottleneck concerns *design formalization*: agents are typically built using heuristic, hard-coded practices that resist auditing and reproducible engineering. Existing methods co-optimize prompt and topology stages through search, with iterative optimization [24], but such labor-intensive optimization remains prohibitively expensive. Emerging views of orchestration and dynamic topologies offer a potential shift toward a broader *utility*-level framework [9]. Model comparisons show that a single-LLM implementation of multi-agent workflows can match homogeneous frameworks while simultaneously reducing inference cost, raising the question of how much the parameterization (e.g., multi-agent versus single-agent) contributes to performance relative to well-known KV-cache reuse [138]. In either case, naive adoption of static roles and fixed communication graphs incurs massive and, more critically, *unpredictable* cost-latency trade-offs across different runs—a crucial deficiency for a production-grade system [47, 48].

Thus, production-scale multi-agent systems must be reframed as a systems engineering problem requiring principled failure analysis, rather than as a pure AI-modeling exercise. The most urgent directions include: (i) the implementation of context-transferable, certificate-verifiable memory and retrieval mechanisms [39]; (ii) bridging the gap between flat logging and framework-neutral evaluation through cross-run analysis, fault-injection, and LLM-transparent computational profiles [139, 23]; and (iii) explicit research on scalability in which pruning, state-awareness, and modularity—rather than a default reliance on agent quantity—drive performance. Accountability in dynamic orchestration and human readability become necessary conditions; otherwise, LLM-based multi-agent systems will remain challenging to adapt, manage, and trust in enterprise settings.

7.5 Paths to Reliability – Security Audit and Evaluation.

As LLM-based multi-agent systems transition from research prototypes to production deployments, the methodologies used to assess their behavior must evolve beyond conventional functional benchmarking toward rigorous security auditing and systematic reliability evaluation. This requires moving from point-in-time capability measurement to continuous, multi-dimensional assessment that encompasses adversarial resilience, systemic failure modes, governance structures, and socio-technical accountability.

The foundational layer of this evaluation agenda concerns the precise identification of emergent attack surfaces. Multi-agent architectures introduce vulnerabilities that fundamentally differ from those in single-agent LLM systems. In contrast to isolated model attacks, compromised agents can exploit inter-agent trust relationships, allowing malicious content to propagate through communication channels beyond the reach of individual defensive mechanisms. The threat model in such systems extends to shared memory infrastructures, where poisoning attacks on common knowledge bases can contaminate interaction trajectories across multiple agents [71]. Furthermore, novel vulnerability classes such as Memory Control Flow Attacks have demonstrated that persistent memory can exert deterministic influence over tool selection and execution paths, enabling attackers to induce behavioral deviations that persist across heterogeneous tasks and interaction horizons [72]. An increasingly critical dimension is the topology-dependence of multi-agent system security, where the precise arrangement and delegation patterns of agents create unique attack opportunities [35]. These findings are complemented by frameworks that systematically characterize cross-domain security challenges—including how a benign agent in isolation can leak secrets or violate policy when interacting with untrusted peers in cross-organizational settings [121].

A second pillar is the analysis of cascade propagation and systemic failures. Multi-agent structures are inherently susceptible to cascading safety issues due to communication and joint decision-making processes, where the introduction of communication channels becomes a significant vulnerability vector [140]. Evaluations under a Hidden Profile paradigm reveal a systematic failure mode distinct from single-agent reasoning: agents fail to recognize latent information asymmetry, prematurely converge on shared evidence, and exacerbate these errors with larger group sizes—a process that undermines the purported scalability advantage of multi-agent settings [141]. High team reliability can be framed in terms of information propagation, ideally suppressing error cascades while preserving beneficial information diffusion [35]. Detection and containment mechanisms are emerging, including optimizing communication structures to prune unreliable messages against adversarial attacks [79] and deploying the monitoring and audit components in evaluating safety.

A critical shift in this roadmap concerns evaluation—conventional static benchmarks fail to infer hidden reasoning or characterize system-level reliability outcomes in the presence of dynamic opportunities for errors and adversarial influence. This is important because detection on static tasks provides limited understanding of system reliability, especially given the evidence of wide variations in performance and robustness across agent architectures [48]. Comprehensive reliability evaluation should incorporate task-specific benchmarks (e.g., latency, failure rates), interaction-based process attestation, and structural failure analysis. Each of these represents an essential effort in its own right. Security audits are integrated into agent-environment workflows themselves—as shown by the study of performance variation, analyses demonstrating that multi-agent architectures are dominant drivers of resource use, reproducibility, cost-latency-accuracy, and even attack outcomes [48]. These empirical foundations provide a calibrated basis for building systematic evaluation frameworks.

From a governance and accountability perspective, formal structures are necessary to define and enforce safety constraints in ways that are auditable. The challenge of Multi-Agent System Routing, where selections of collaboration modes and individual LLM routes directly determine quality and cost trade-offs, requires governance from a protocol strategy to ensure [25]. Addressing through an orchestration methodology offers a structural view to assess reliability: decoding control mechanism that governs message flow, contextual acquisition, and memory management; such a "collaborative memory framework" that presents a protocol for access control and auditing [87]. Similarly, approaches architecture that draws inspiration from operating system design introduces scheduling to adaptively allocate computational resources and memory contexts, addressing several kinds of failures in multi-turn deployment as incidental delay and context degradation. These interfaces are the foundation for systematic and security-aware governance both in inference and in monitoring [50].

The final dimension is human-AI oversight. A key future direction is the development of human-in-the-loop methodologies that align security audits with socio-technical systematic risk, ensuring that the mechanisms designed to ensure reliable behavior are themselves grounded in the actual constraints of the pipeline [51]. This requires tracing the performance and reliability of individual LLMs to the entire agent structure, while accounting for the effects of externalized memory of review: persistent state, reusable skills, interaction protocols, and transferable knowledge [31]. The long-term consequence is that reliability is not a static property measured once, but a continuous property of the system to be governed across its lifecycle. A holistic integration of threat modeling, mechanism oversight, and pervasive monitoring is essential to deploy multi-agent systems with robustness and trust.

7.6 Societal and Ethical Implications, and Long-term Research Agenda

The proliferation of LLM-based multi-agent systems (MAS) marks a fundamental transition in how artificial intelligence is deployed—from isolated tools to interconnected digital societies capable of autonomous coordination, negotiation, and collective action. As these systems move

from research prototypes to production deployments across software engineering, scientific discovery, financial systems, and social simulation [16, 142], the societal and ethical dimensions of the reliability challenges identified in the previous section demand systematic examination. The evaluation agenda established—spanning attack surface identification, cascade analysis, and governance infrastructure—provides the technical vocabulary needed to articulate and confront the broader implications that arise when these systems become socially embedded.

A central concern in this ethical assessment is the amplification of information integrity threats, which directly instantiates the vulnerability classes identified earlier. Multi-agent architectures introduce qualitatively new attack surfaces: adversarial content can propagate through inter-agent communication channels, shared memory stores can be poisoned, and malicious payloads can cascade across trust boundaries—all phenomena surfaced as systemic risks in the preceding reliability analysis [104]. Evidence indicates that models which resist direct attacks are nonetheless compromised when requests originate from peer agents, with nearly all tested LLMs executing malicious content under these conditions. The structural properties of agent networks—their topology, communication protocols, and information routing—thus become first-class risk factors that shape both capability and vulnerability [35]. The "Telephone Loop" attack further demonstrates that cross-agent delegation cycles can degrade performance in ways single-agent architectures are immune to [7], underscoring how structural interdependencies compound individual weaknesses. As multi-agent systems scale from dozens to thousands of nodes, the potential for cascading failures and systemic misinformation propagates beyond isolated organizations, threatening the integrity of the information ecosystem these systems increasingly inhabit.

The economic and labor implications of large-scale agent deployment warrant equally serious attention. The transition toward multi-agent autonomy raises fundamental questions about which cognitive functions remain with humans, which are delegated to machine collectives, and how value creation and destruction are distributed across the workforce [143]. Multi-agent systems that solve million-step tasks with zero errors [144] and orchestrate complex workflows optimize resource-allocation problems fundamentally different from those of single-agent deployments, altering who benefits from productivity gains and who bears the costs. When the economic significance of such systems matures, they simultaneously threaten to concentrate power among those who help control agent infrastructure and to reshape human roles in knowledge-intensive occupations. This concentration of control—over central orchestrators, access to high-quality models, and the data ecosystems that shape system behavior—creates new vectors for economic inequality. Effective regulation must, therefore, replace the unit of analysis from individual algorithms to populations of interacting agents, with accountability distributed across networks rather than isolated to model outputs. A new legal vocabulary is needed to allocate responsibility when agents act collaboratively—and when they fail collectively—thus distinguishing robust systemic oversight from reparations for component-level malfunction.

The simulation of human and social dynamics, while methodologically valuable for computational social science [145], raises deeper ethical questions about digital doubles, consent, and the purposes of manufactured collectives. When researchers create societies of language model agents to model policy responses or render complex social behavior, the boundary between simulation and reality, between observation and intervention, becomes both diagnostic and philosophically salient [146]. These simulations are not mere models—they interact with small group dynamics and emergent ecological shifts with our evolving ecosystems [147]. The risk is a double form of capture: not only constraining the discipline that can be respected, but also normalizing the notion that social behavior can and should be regulated, transforming unpredictability into a failure mode to be optimized away.

A coherent long-term research agenda must thereby embed ethical reasoning directly into generative design frameworks. The same automated optimization machinery that searches over agent prompts, topologies, and coordination strategies [24, 148] must also optimize for safety

guarantees, intervention points, and resilience against adversarial perturbations. This goes hand-in-hand with the governance infrastructure discussed previously, accenting, beyond auditability, that safety requirements must be integrated into design objectives. Programmatic approaches can improve the reliability and observability of such systems [48, 82], offering mechanisms for systematically searching for failures before deployment. The RACI framework [149] offers one pathway to co-designing oversight that preserves accountability while retaining the benefits of distributed autonomy.

Long-term research must further embrace a resilience-centered perspective, rather than treating robustness as the mere elimination of known faults. This includes developing self-repairing architectures that dynamically reconfigure communication topologies in response to detected compromised agents, establishing unified benchmarking frameworks that incorporate vulnerability assessment into the standard design loop, and building governance structures where security is a first-class design constraint rather than an afterthought. The increasingly sophisticated capacity of these systems to coordinate over multi-turn horizons underscores the urgent demand for realistic benchmarks that measure resilience beyond single-step accuracy [136]. In investigating the circumstances under which information flows across agent graphs collapse into monocultures, redundancies, or error cascades [35], the research community must encode resilience as a foundational principle of agent design. Without a forward-looking research agenda—armed with realistic benchmarks, multi-scale evaluation, and accountable governance—the emergence of communities of AI agents could constitute an unaccountable digital ecosystem, with concentrated risk for those excluded from its development and governance.

8 Conclusion

This survey has charted the intellectual trajectory of large language model-based multi-agent systems, tracing their evolution from early conceptual antecedents in distributed artificial intelligence and classical multi-agent reinforcement learning to the contemporary paradigm of collaborative LLM collectives [5, 4]. The field has matured considerably: the theoretical impetus for decomposing problem-solving across multiple interacting agents is now substantiated by systematic studies demonstrating that heterogeneous configurations can yield substantial performance gains over homogeneous scaling [58], and research has verified that diverse LLM-driven teams harness distinctive collective potential [56, 19]. Concurrently, foundational investigations in emergent coordination have established that prompt-level interventions—such as assigning personas and inducing theory-of-mind reasoning—can steer aggregate agent groups toward higher-order collectives characterized by goal-directed complementarity and identity-linked differentiation [81]. This capacity for continuous evolution is a central theme of the field: multi-agent systems are not static architectures but dynamically adapting organizations, capable of self-improvement through iterative adaptation and co-evolution. Co-evolutionary reinforcement learning frameworks where proposer, solver and critic agents mutually reinforce each other’s capabilities [91], and decentralized, graph-based autonomous coordination frameworks that maintain dynamic task routing and local expertise [11], demonstrate this capacity. Furthermore, specialized orchestration models can automatically discover powerful coordination strategies among LLMs and adaptively sequence agents in real time, generating effective communication topologies and optimizing task allocation from pure end-to-end reward maximization [60]. The capacity for automated systems design has expanded the design space for MAS, demonstrating prompt and topology co-optimization [150, 24], and at an even broader level, facilitating query-dependent sampling from an agentic supernet [46]. These advances, however, are not without rigorous cautionary findings: our investigation of the actual performance and reliability of MAS across reproducible settings has exposed a disproportionate focus on the latest outcomes over a faithful evaluation of the underlying system. Comprehensive dataset analysis reveals failures arising from inter-agent misalignment and task-verification vacuums, among other causes

[7]. MAESTRO additionally exposes substantial run-to-run variance in the performance and reliability of popular frameworks, such that structural and temporal variability of executions can yield divergent conclusions [48].

A central tension emerges from these collective efforts: the mechanism of multi-agent architecture—which excels at decomposing and delegating tasks across the communication hierarchy—does not scale uniformly in performance. Broad, controlled evaluations across canonical architectures reveal a counter-intuitive effect: coordination yields diminishing returns once single-agent accuracy exceeds a capability threshold; architectural alignment matters substantially, and mismatched coordination can propagate errors and degrade performance while, on decomposable reasoning tasks, collaborating systems significantly outperform their single-agent baselines [13]. These observations resonate with simulation-based analyses indicating that self-organizing multi-agent teams often fail to leverage their expert agent’s performance and, instead, average expert and non-expert views, thus harming the resulting solution [10]. In parallel, distributed agent networks generate new challenges—such as managing the coordination overhead of communication and context [8], and facing novel security threats, where vulnerabilities arise in the system and through inter-agent communication and tool integration [104, 104]. These concerns amplify the need for rigorous, standardized evaluation frameworks capable of systematically assessing reliability, reproducibility, and interaction quality to guide future design [23, 47]. At the system-design interface, the capacity resource-aware allocation, dynamic topology selection, and efficient orchestration—the dual imperative to optimize performance while reducing cost and latency—is material and in the architecture frontier [25, 23, 151].

In conclusion, multi-agent LLM systems mark a fundamental re-framing of AI’s unit of operation, shifting from a single fixed model to an ecosystem of coordinated, continually adapting cognition [33]. The most consequential future advancement will be to both refine the individual agent as well as to give architectures the capacity to learn and evolve as a collective organizational unit [91, 88]. The promise of cognitive super-additivity—where group performance exceeds the sum of flawed parts—coexists with perils of cascading error, security vulnerability, and misalignment with human values [10, 77]. The field’s next milestone—and its unquestionable intellectual frontier—depends on formalizing the "multi-agent scaling law" into reliable, robust and disciplined design principles that unify foundational architectural awareness, communication, and an adaptive orchestration mechanisms into a trustworthy theory of collective decision-making [13, 14]. The convergence of these collective intelligence architectures with computational social science, as explored in studies of emergent group dynamics and social phenomena, further scenarios underscores the most promising horizon: the aggregation and orchestration of human-like intelligences with the robustness and principle that will eventually shape the future of LLM-based, socially embedded AI [22, 80].

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